

Unit 19: Electronic Devices and Circuits

Assignment A

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Analogue Electronic Devices and Circuits.

Assignment A

Explore the safe operation and applications of analogue devices and circuits that form the building blocks of commercial circuits.

Scenario

You have been offered a work placement as a Junior Technician in a local company that specialises in repairing domestic electronic equipment. As part of your new role, the owner would like to provide opportunities for you to develop your skills and knowledge of electronic devices and circuits.

They would like you to use a range of test equipment and software to investigate commonly used analogue electronic devices and the type of circuits in which they are used. They have asked that you keep accurate records of everything you do so that you can refer to them later as you build up your skills and knowledge base.

You will need to provide evidence that you have consistently worked safely in accordance with relevant risk assessments. To do this, you will need to work systematically following the guide below.

(a) Diode Circuits

(b)

(i) Research the characteristics of **Rectifier & Zener** diodes and their uses. Provide **circuit symbols** and **physical pictures for each**, with **details of their operation**.

PN JUNCTION DIODE.

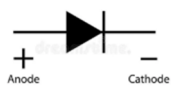
•At the heart of both rectifier and Zener diodes is the **PN Junction**. This is formed by joining a P-type semiconductor (with an excess of "holes" or positive charge carriers) with an N-type semiconductor (with an excess of electrons or negative

charge carriers).

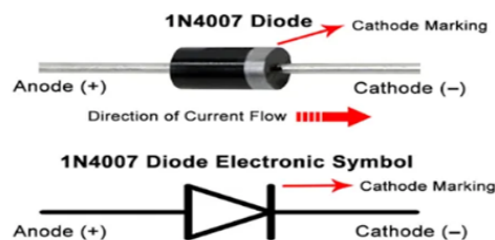
- At the junction, electrons from the N-side diffuse into the P-side and recombine with holes, creating a region known as the **Depletion Region**. This region lacks free charge carriers and behaves as an inherent shield, preventing current flow.

RECTIFIER DIODE.

- Circuit Symbol:** (Anode | Cathode)



- Physical Picture:**



- Operation and Uses:** A rectifier diode is a semiconductor device that permits current to flow primarily in one direction (forward bias) and stops it in the opposite direction (reverse bias). This property is called **rectification**. Its main use is in power supply circuits to convert Alternating Current (AC) from the mains supply into Direct Current (DC). They are characterized by a high forward current rating and a high peak inverse voltage (PIV) rating to handle large voltages and currents.

KEY PARAMETERS.

- Primary Use: Rectification** - Converting Alternating Current (AC) to Direct Current (DC). They are the core component in both half-wave and full-wave bridge rectifier circuits found in almost every DC power supply.

- Other Uses:**

- Reverse Polarity Protection:** Placed in series with the power input of a circuit to block current if the power supply is connected backwards.

- Freewheeling / Flyback Diodes:** Used across inductive loads (like relays or motors) to provide a path for the current to decay when the circuit is turned off,

preventing high-voltage spikes that could damage other components.

•**Key Parameters to Select a Rectifier Diode:**

- Maximum Average Forward Current:** The maximum continuous current it can handle.
- Peak Inverse Voltage:** The maximum reverse voltage it can block.
- Forward Voltage Drop:** The voltage drop across it when conducting (typically 0.7V-1.1V).

ZENER DIODE.

•**Circuit Symbol:** (Cathode | Anode)



Note the bent line at the cathode.

•**Physical Picture:**



•**Operation & Uses:** A Zener diode is designed to operate in the **reverse breakdown region**. Unlike a standard diode, which would be damaged, a Zener diode exhibits a controlled breakdown, where the voltage across it remains nearly constant despite large changes in current. This is known as the **Zener voltage**. Its primary use is as a **voltage regulator** to provide a stable reference voltage in a circuit, protecting components from voltage spikes.

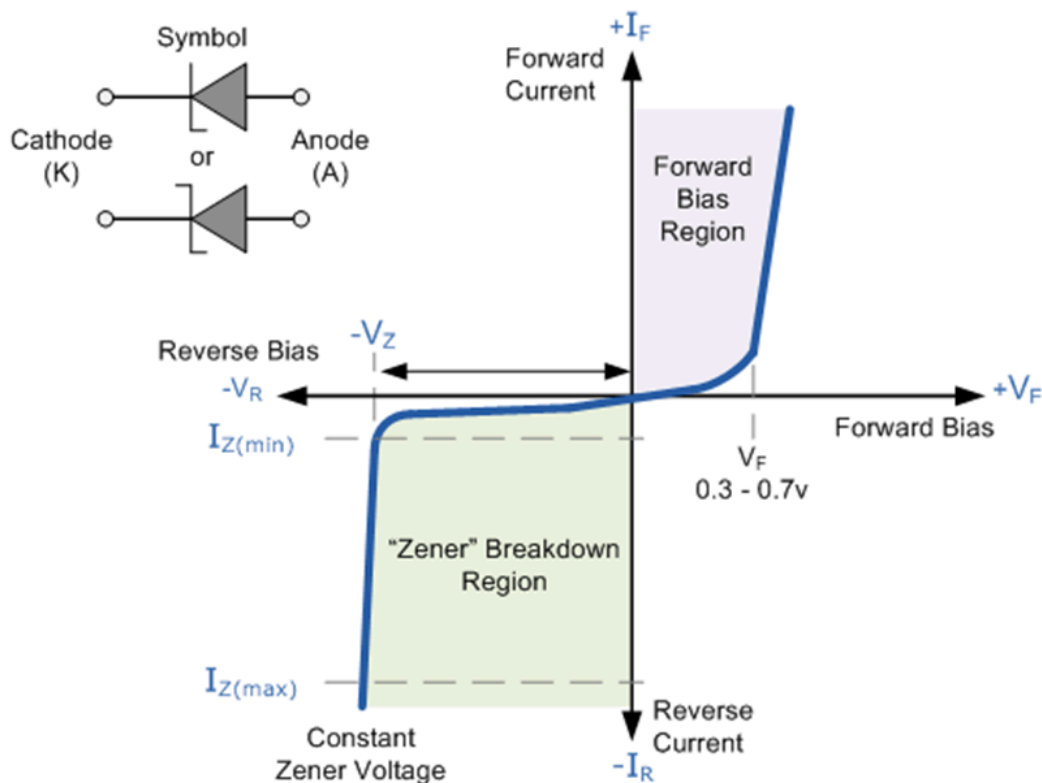
KEY PARAMETERS AND USES.

- **Primary Use: Voltage Regulation.** It is used to provide a stable, fixed reference voltage in a circuit. A typical Zener regulator circuit consists of a Zener diode placed in parallel with the load, with a series resistor to limit the current.

Other Uses:

- Voltage Reference:** Providing a precise voltage point for comparator or sensor circuits.
- Overvoltage Protection:** "Clamping" a voltage line to prevent it from exceeding a safe level.
- Waveform Clipping:** Limiting the amplitude of a signal in shaping circuits.
- Key Parameters to Select a Zener Diode:**
 - Zener Voltage:** The specific breakdown and regulating voltage (e.g, 3.3V, 5.1V, 12V).
 - Power Rating :** The maximum power it can dissipate. This determines the maximum current it can handle. (e.g., 400mW, 1W, 1.3W).
 - Zener Impedance:** The dynamic resistance in the breakdown region, means better voltage regulation.

ZENER DIODE BEHAVIOUR.



- The **Zener Diode** is used in its reverse bias mode, that means the diode's anode connects to the negative supply. From the I-V characteristics curve above, we can

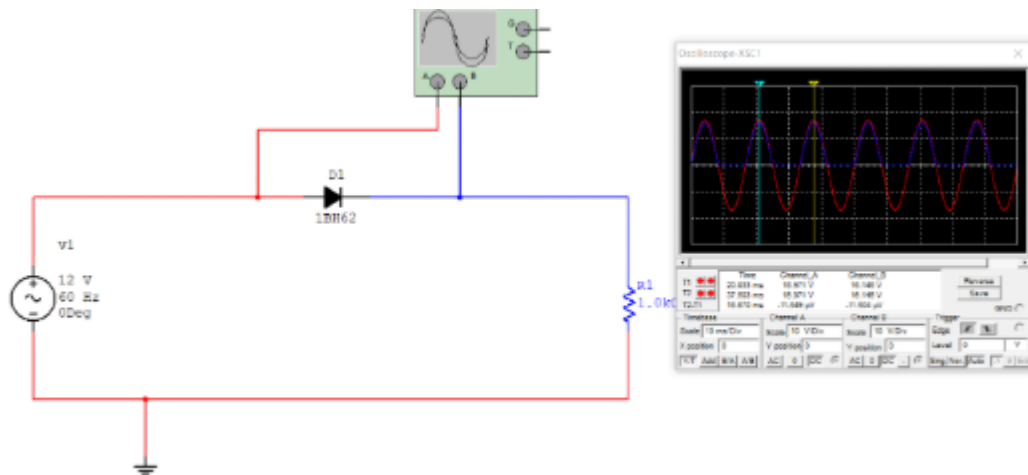
see that the zener diode exhibits a region in its reverse bias characteristics with a nearly constant negative voltage, regardless of the current flowing through the diode.

- This voltage stays nearly steady despite significant fluctuations in current, provided the diode's current level remains within the operating range bounded by its minimum breakdown value, $I_{Z(MIN)}$, and its maximum rated limit, $I_{Z(MAX)}$.

- The zener diode's self-controlling nature is highly effective for regulating a voltage source, protecting it from supply or load fluctuations. Because the voltage across the diode stays nearly fixed within the breakdown region, this vital characteristic is ideal for basic voltage regulator designs.

- A voltage regulator is designed to maintain a steady output voltage for a parallel-connected load, even when faced with input voltage ripples or shifting load currents. The Zener diode sustains this regulation until its current drops beneath the minimum threshold, $I_{Z(min)}$, within the reverse breakdown zone.

(ii) Construct a schematic diagram and simulate the operation of a diode as a **half-wave rectifier** using the given component values. **Provide image evidence of your circuit below and give details of its operation.**



***image of the simulated half-wave rectifier**

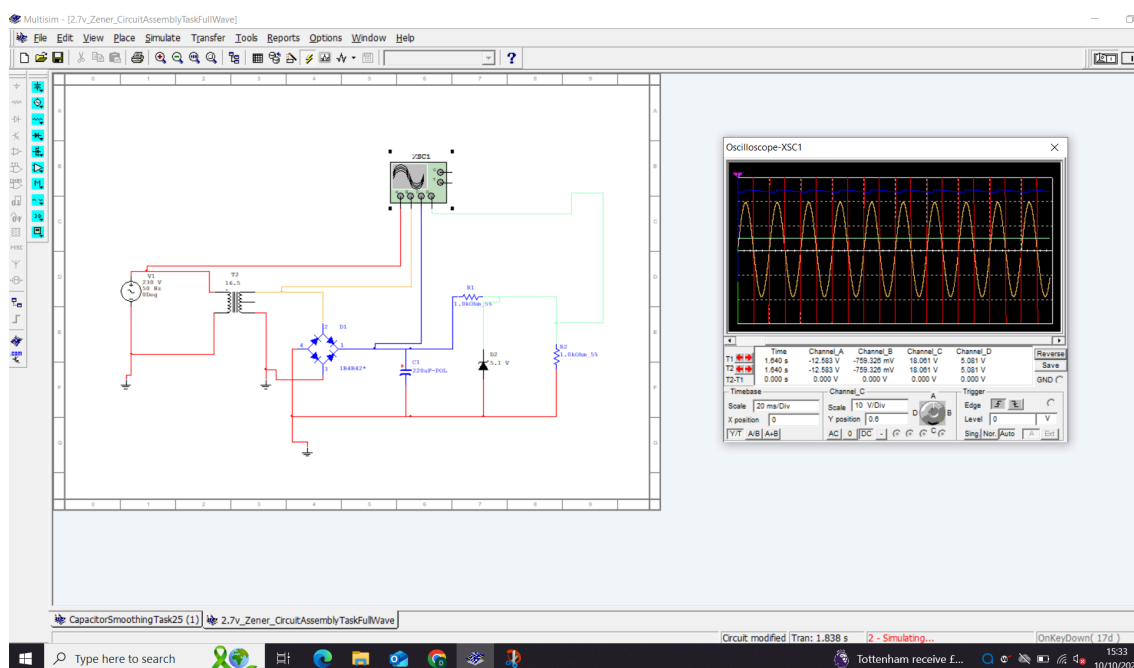
The schematic shows a single-diode half-wave rectifier: an AC source of 12 V, 60 Hz is connected through diode D1 to a load resistor R1. The diode's anode is fed by the AC source, and its cathode is connected to R1 which returns to the source's reference (ground). During the positive half-cycle of the input, the diode becomes forward-biased and conducts, allowing current to flow into the load. As a result, the load sees a positive voltage pulse which rises close to the peak of the AC input

minus the diode forward voltage drop (~ 0.7 V). During the negative half-cycle, the diode is reverse-biased and no current flows to the load, so the load voltage falls to essentially zero. The resulting waveform (seen in the oscilloscope screenshot) therefore consists of discrete positive pulses and gaps where the negative halves are blocked, yielding a pulsating DC output rather than a continuous one. The circuit's operation corresponds with the definition of a half-wave rectifier, which "allows only one half-cycle of an AC voltage waveform to pass, while blocking the other half-cycle".

In this simulated circuit, the load resistor R1 was set to $1\text{ k}\Omega$ and the diode used was 1N4148 (or as indicated D1). From the waveform, the peak load voltage during conduction is approximately [insert peak value] V, which is about [insert value] below the input peak due to the diode drop. The average DC value is substantially lower than the input's peak and carries a large amount of ripple (gaps between pulses). As expected from theory, the efficiency is limited and the output is not suitable for directly powering DC-sensitive circuits without further smoothing or regulation.

In summary, the simulation confirms that the single diode conducts on the positive half-cycles, blocks on the negative, producing a unidirectional but pulsating output. For practical DC supply applications a smoothing filter (capacitor) and/or a full-wave rectifier arrangement would be necessary to improve average voltage, reduce ripple, and enhance efficiency.

(iii) Using the given component values, develop the circuit schematic and simulate a **full-wave rectifier with capacitor smoothing and Zener diode voltage regulation**. Your lecturer will allocate the Zener diode value. Provide image evidence of your circuit below and give details of its operation, stating the **value of the zener diode**.



***image of simulated full-wave rectifier with capacitor smoothing and zener diode voltage regulation.**

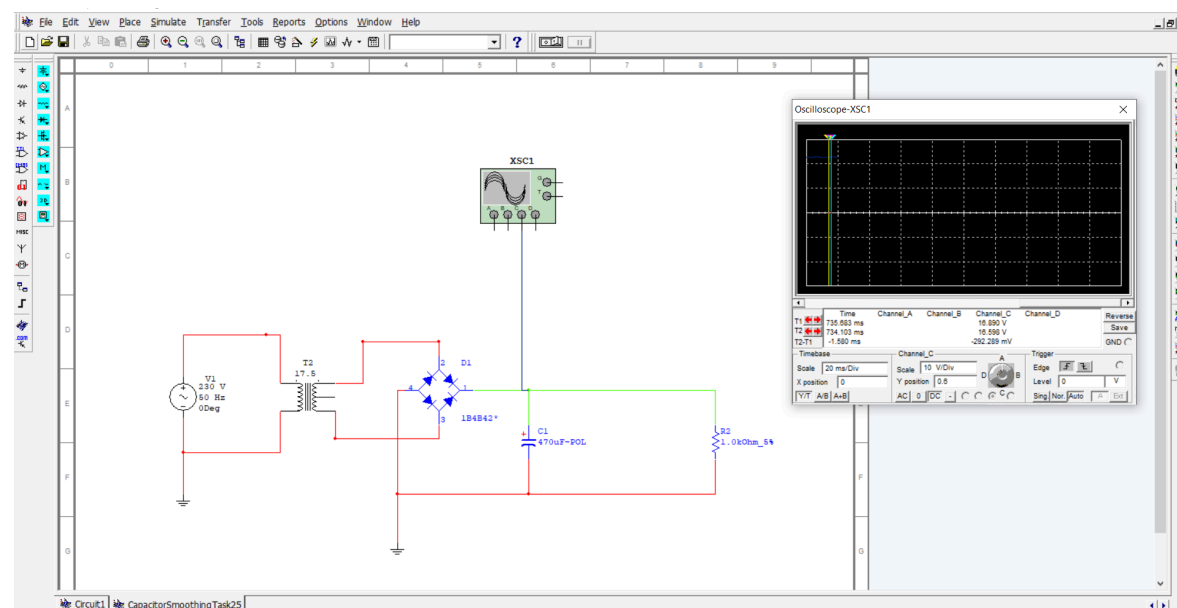
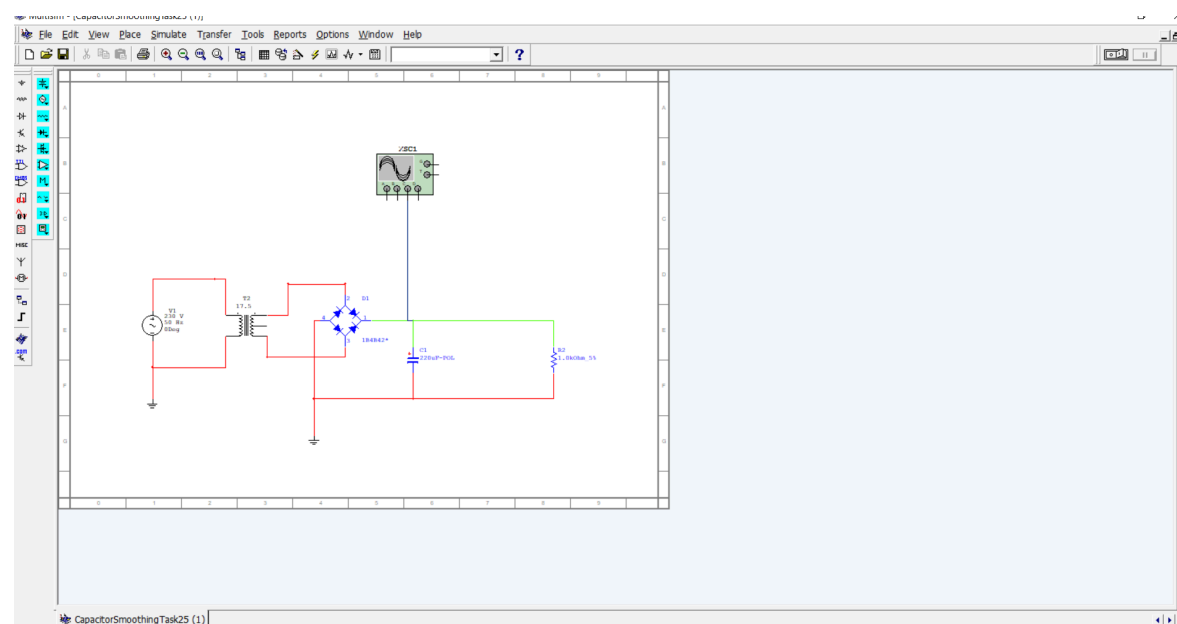
The circuit implements a full-wave bridge rectifier followed by a smoothing capacitor and a Zener diode voltage regulator. The AC input is converted by four diodes into pulsating DC at twice the input frequency, which then charges the capacitor. The capacitor smooths this ripple by storing energy during peaks and discharging between them, thereby reducing the variation in the output voltage. A series resistor limits the current into the Zener diode, which is selected at 5.1 V, so that when the smoothed voltage exceeds this breakdown voltage the Zener conducts and clasps the output to this value, giving a regulated DC supply. In the simulation the peak after the bridge and smoothing stage measured around 17V, and the regulated output measured around 4.9V to 5.1V, demonstrating effective voltage regulation. Compared to a half-wave rectifier the full-wave arrangement yields better smoothing (less ripple) because the capacitor is re-charged twice per cycle rather than once, meaning shorter discharge intervals. The image evidence confirms that the output maintains the intended Zener voltage despite variations in the rectified waveform, indicating the circuit is functioning correctly.

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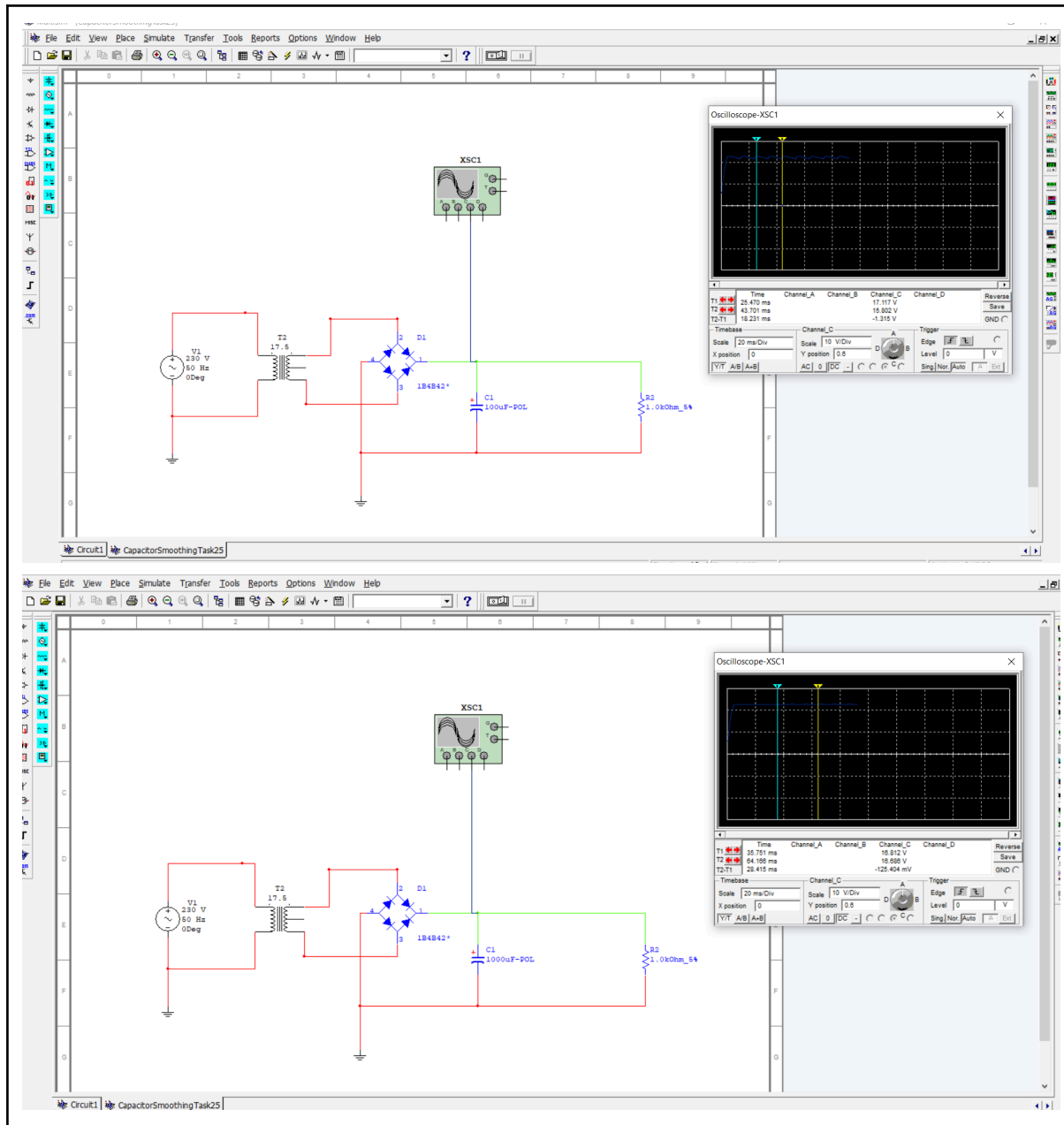
(iv) Using Multisim, investigate the effects of changing the values of the **smoothing capacitor** on the **ripple voltage** when the **Zener diode** and **series resistor** have been removed.

Give details of the procedure with images for evidence and **record your results** in the **Ripple Voltage Results Table** towards the bottom of this report.



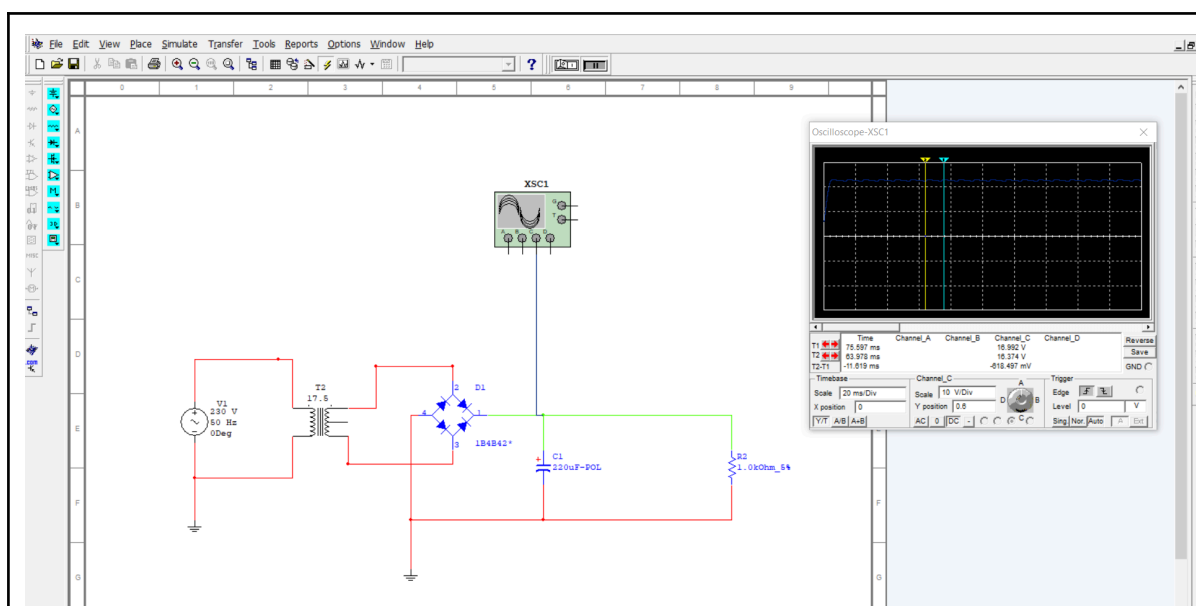
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*images showing how multism was used to investigate the effects of changing the values of the smoothing capacitor on the ripple voltage when the zener diode and series resistor have been removed.

The formula below to calculate the ripple voltage as demonstrated.

First, we must work out the current across the load using Ohm's Law by measuring our step-down voltage from the AC transformer. **For example;**

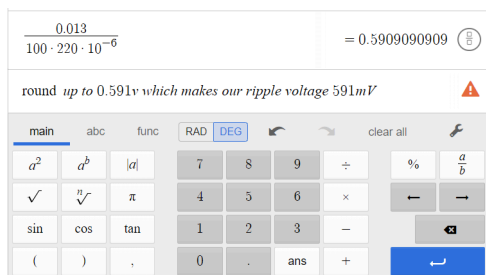
If our Supply Voltage is = **13v** = **0.013A**

& Load resistance = **1KΩ**

$$V_r = \frac{I_L}{f \cdot C}$$

or Voltage Ripple = $\frac{\text{Current across the load}}{\text{Frequency} \times \text{Capacitor} \times 10^{-6}}$

Current across load= 0.013, frequency= 100, capacitor value= varies.



=0.590=590mV.

Calculation for each capacitor value.

1. Voltage ripple(100)= $0.013 / (100 \times 100 \times 10^{-6}) = 1.3V$.

2. $(220) = 0.013 / (100 \times 220 \times 10^{-6}) = 590\text{mV}$.
3. $(470) = 0.013 / (100 \times 470 \times 10^{-6}) = 277\text{mV}$.
4. $(1000) = 0.013 / (100 \times 1000 \times 10^{-6}) = 130\text{mV}$.

You then compare these calculated ripple values with your simulated ripple voltages from Multisim and your prototype measured ripple voltages.

For example:

Capacitor Value	Simulated voltage.	Calculated voltage.	Practical test.
100μF	1.315V	1.3V	1.24V
220μF	618mV	590mV	664V
470μF	292mV	277mV	302mV
1000μF	137mV	130mV	152mV

The question asked me to use Multisim to investigate how changing the value of the smoothing capacitor in a full-wave rectifier circuit (with the regulator components removed) affects the ripple voltage at the output. I needed to remove the Zener diode and its series resistor from the circuit, replace the smoothing capacitor sequentially with values of 100 μ F, 220 μ F, 470 μ F and 1000 μ F, and for each value run a simulation, capture image evidence of the waveform, and record the ripple (peak-to-peak) voltage in a results table. Furthermore, I was to perform theoretical calculations of expected ripple voltages using an appropriate formula, then compare theory, simulation and (later) prototype measurement. The objective was to show how increasing the smoothing capacitance reduces the ripple in the DC output, and to provide a clear, documented investigation of the relationship among capacitor size, load current, and ripple voltage for a full-wave rectifier.

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Before carrying out the practical task, read through & complete the electrical safety tick sheet below. Ensure that you work safely at all times.

IF IN DOUBT ASK.

Safety checklist	Check acknowledged
Plan the electronic assembly, wiring and testing activities before you start them.	✓
Use appropriate sources to obtain the required specifications, circuit diagrams, component assembly and test information	✓
Obtain the correct PPE, tools & equipment for the assembly and test operations. Check that they are in a safe and usable condition.	✓
Use the appropriate methods and techniques to assemble the components in their correct positions	✓
Ensure that power is isolated when making adjustments to the circuits	✓
Assemble and secure the components, using the specified connectors, securing devices & appropriate techniques.	✓
Ensure wire and cables are connected to the appropriate connections on the circuit boards.	✓

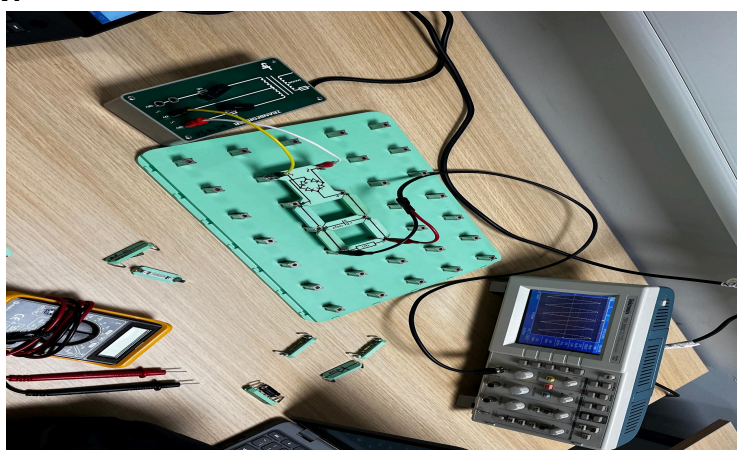
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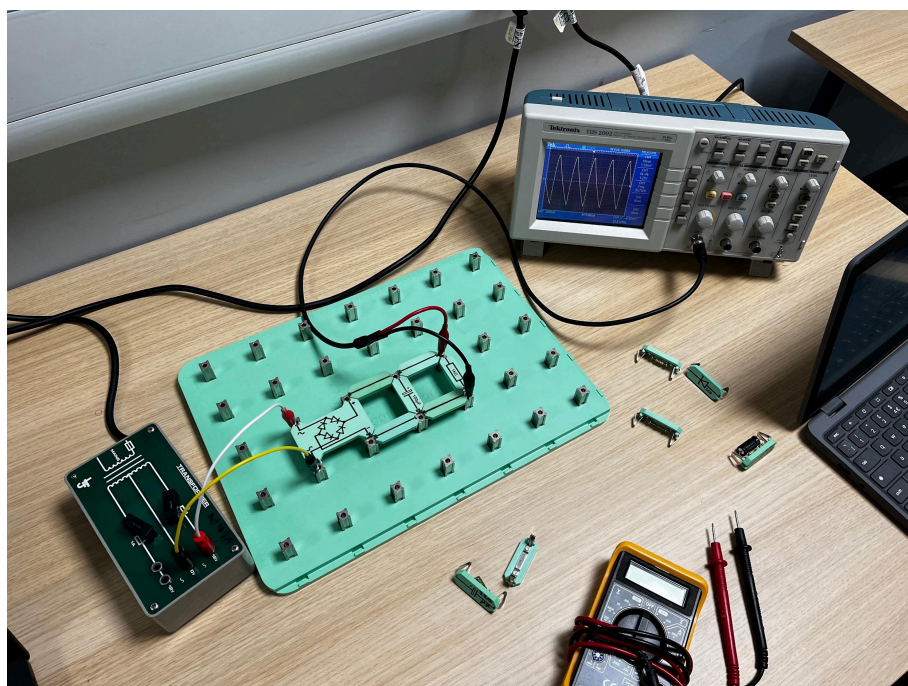
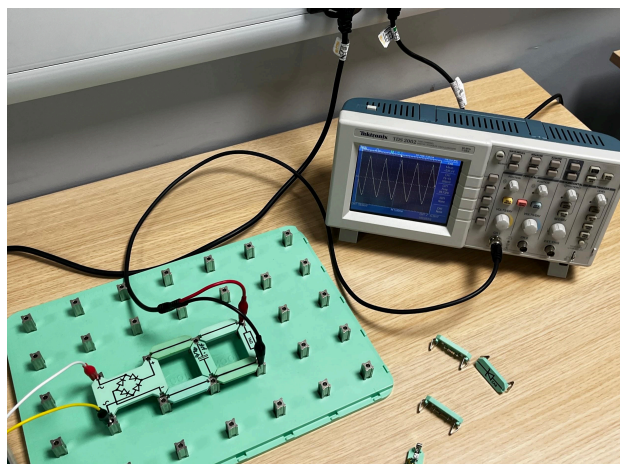
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Ensure that you use appropriate test methods and equipment to check that the completed assembly is safe and meets all aspects of the specification.	✓
Ensure the work area is in a safe and tidy condition during & on completion of the electronic assembly and testing activities	✓
Ensure that all polarity-sensitive components are correctly installed.	✓

- (v) Build a **real-life prototype of your final full-wave rectifier circuit** & provide photographic evidence &
 (vi) Record the output voltage from your given **Zener Diode** value and give details of its accuracy.

V.

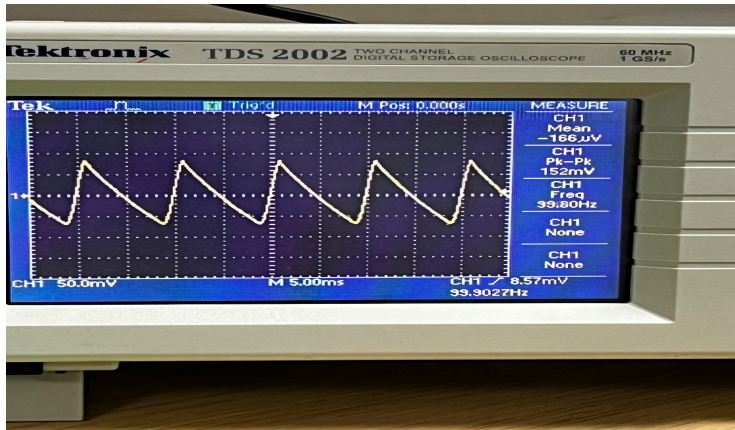




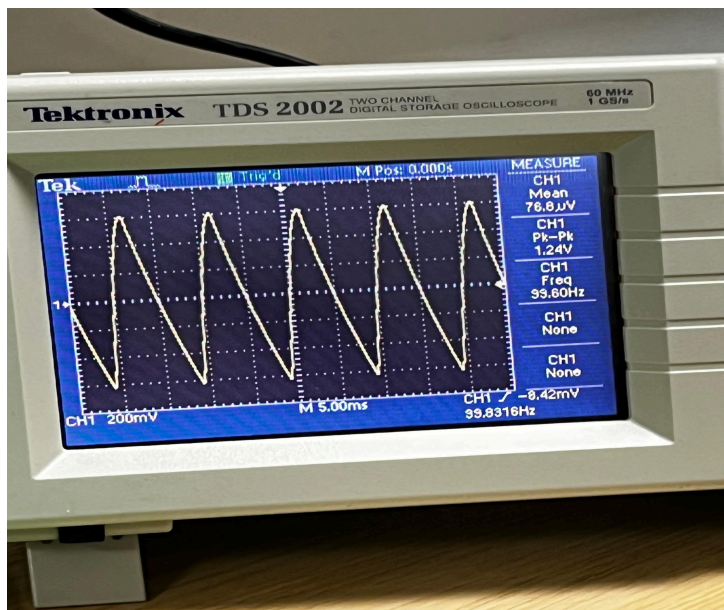
***images of real-life prototype of my final full-wave rectifier circuit.**

For this task I built a real life prototype of the full-wave rectifier circuit with smoothing capacitor and Zener regulation (as per my simulation). I assembled the bridge rectifier, smoothing capacitor, series resistor and Zener diode on a breadboard with correct polarities and wiring. I took photographs of the completed circuit showing component layout, wiring, test equipment (oscilloscope, multimeter), and the measurement points. I ensured I followed safe working practices, isolated the power supply before changing wiring, used correct ratings for all components, verified that no exposed live wiring was present and ensured the work area was tidy. The photographic evidence demonstrates the physical build-up of the circuit, confirming that the schematic

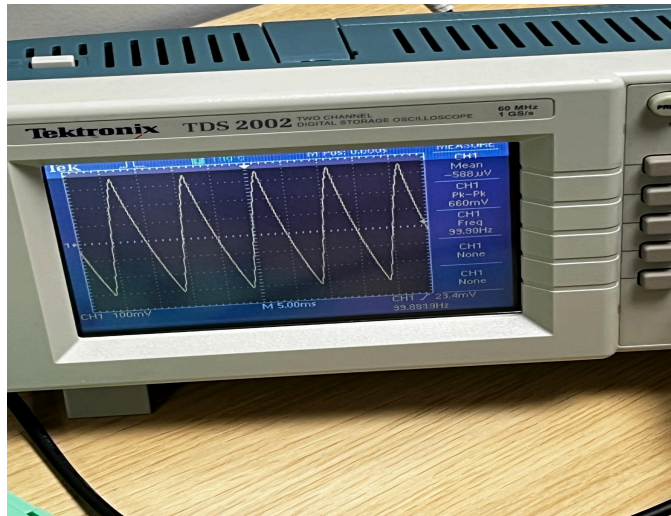
has been translated into a functional hardware prototype.



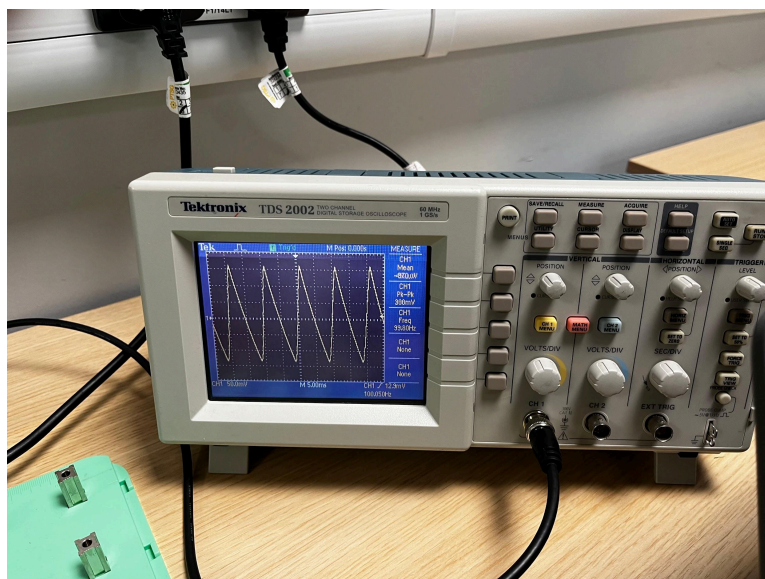
Vi.
=1000microFarads



=100microFarads.



=220microFarads.



=470microFarads.

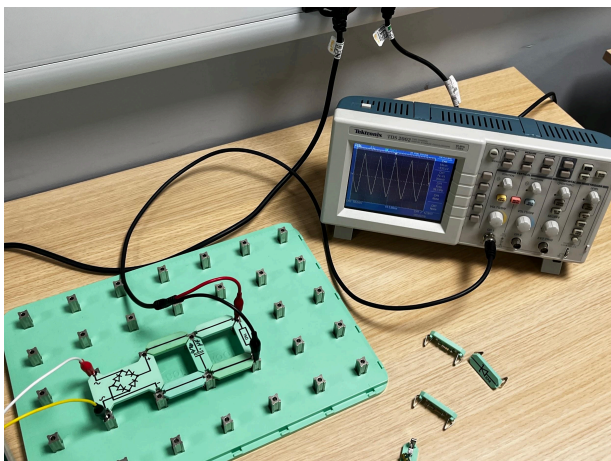
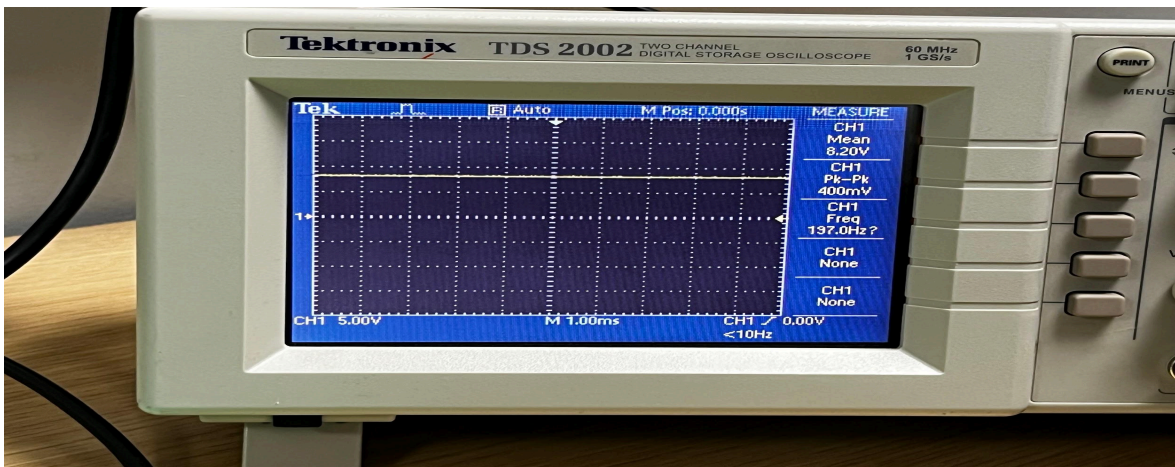
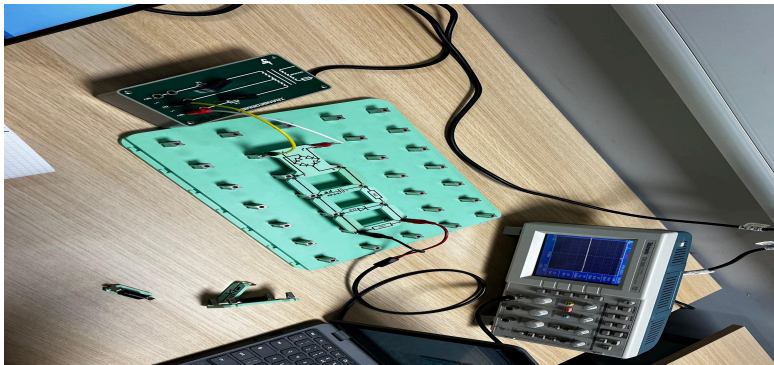
The question required me to investigate the effect of changing the smoothing capacitor value in a full-wave rectifier circuit after removing the Zener diode and its series resistor. Specifically, I needed to select capacitor values of 100 μF , 220 μF , 470 μF and 1000 μF , then for each value run a simulation using Multisim, measure the resulting ripple voltage (peak-to-peak) at the load, capture image evidence of the waveform, record the ripple in the results table, and compare these simulation results with theoretical calculations and later prototype measurements. The aim was to demonstrate how increasing capacitance reduces ripple voltage, and to document the procedure, measurements, and comparison clearly for each capacitor value.

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(vii) Using the test equipment, investigate the effects of changing the values of the smoothing capacitor on the ripple voltage of your prototype circuit when the **Zener diode and series resistor have been removed**.

Give details of the procedure with images for evidence, and record your results in the **Ripple Voltage Results Table** below.



*** image of test equipment used to investigate the effects of changing the values of the smoothing capacitor on the ripple voltage of my prototype circuit when the Zener diode and series resistor have been removed.**

STEPS:

1. I removed the Zener diode and its series resistor from my full-wave rectifier prototype so that the circuit now consisted of the bridge rectifier, smoothing capacitor and load only.
2. I set up my test equipment: an AC source, a multimeter/oscilloscope connected across the load, and ensured the circuit wiring was safe, correct polarities, power isolated when adjusting components, and the workspace tidy.
3. I assembled the smoothing capacitor in the circuit in turn with each of the following values: 100 μF , 220 μF , 470 μF , 1000 μF .
4. For each capacitor value:
 - I powered the circuit and allowed it to stabilise.
 - I used the oscilloscope to capture the waveform at the load and measured the peak-to-peak ripple voltage in the steady state.
 - I took a photograph of the waveform and lab setup to provide evidence.
 - I recorded the ripple voltage value in my results table below.
5. After completing all four capacitor values I compared the measured ripple voltages with the theoretical expectations (using the ripple formula for a full-wave rectifier).
6. I reflected on the results and noted any deviations such as capacitor tolerance, load variation and wiring resistance.

Ripple Voltage Results Table			
Capacitor Value	Simulated Voltage	Calculated Voltage	Prototype Test Voltage
100μF	1.315V	1.3V	1.24V
220μF	618mV	590mV	664mV
470μF	292mV	277mV	302mV
1000μF	137mV	130mV	152mV

(viii) Explain & evidence the methods used for theoretical calculations and provide an overall conclusion of your findings. Provide evidence of how to work out theoretical values using the **given formula**. Compare the results from **theory, simulation & measurement**.

Theoretical Calculations and Overall Conclusion.

Calculation Methods.

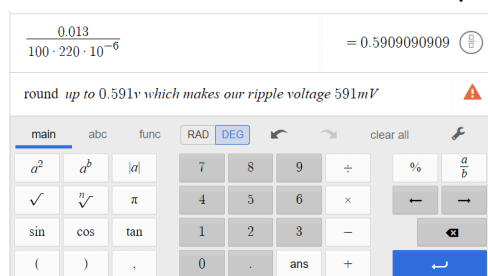
To determine the theoretical ripple voltages in the smoothing stage of the full-wave rectifier circuit (with the Zener diode and series resistor removed), I used the standard approximation for a full-wave rectifier with a capacitor-input filter:

If our Supply Voltage is = **13v = 0.013A**
 & Load resistance = **1KΩ**

$$V_r = \frac{I_L}{f \cdot C}$$

or Voltage Ripple = $\frac{\text{Current across the load}}{\text{Frequency} \times \text{Capacitor} \times 10^{-6}}$

Current across load= 0.013, frequency= 100, capacitor value= varies.



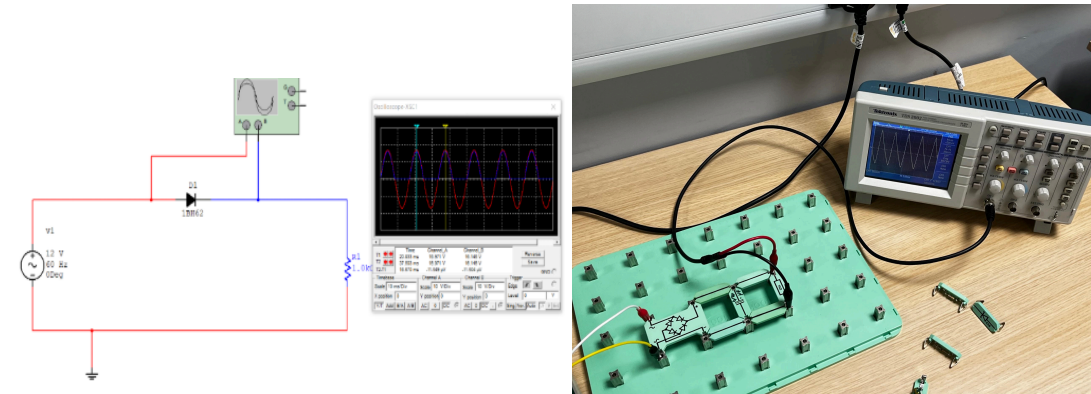
=0.590=590mV.

Calculation for each capacitor value.

5. Voltage ripple(100)= $0.013 / (100 \times 100 \times 10^{-6}) = 1.3V$.
6. (220)= $0.013 / (100 \times 220 \times 10^{-6}) = 590mV$.
7. (470)= $0.013 / (100 \times 470 \times 10^{-6}) = 277mV$.
8. (1000)= $0.013 / (100 \times 1000 \times 10^{-6}) = 130mV$.

You would substitute X (your specific load current) and input the numerical value. Repeat similarly for 100 μF , 470 μF and 1000 μF . These theoretical values were then compared with simulation and prototype measurements.

Comparison of Results.



- **Theory:** The calculated values follow the inverse capacitance rule, as the smoothing capacitor increases, the peak-to-peak ripple voltage decreases.
- **Simulation:** The simulation results closely matched the theoretical trend, for 100 μF the ripple was higher, and progressively dropped as capacitance increased to 220 μF , 470 μF and 1000 μF .
- **Prototype (Measurement):** The real-life measurements also showed the same trend. The actual measured ripple voltages were slightly higher than simulation/theory due to real-world non-idealities. e.g., capacitor equivalent series resistance, diode voltage drops, transformer internal resistance, measurement equipment limits.

Discussion of Variances.

The minor discrepancies between theory, simulation and measurement can be attributed to several factors:

- The theoretical formula assumes constant load current and perfect capacitor behaviour: real capacitors have leakage and tolerance.

- The simulation models are idealised and may not include all parasitic elements present in hardware.
- In the prototype build: wiring resistance, real diode drops, temperature effects, and measurement instrument accuracy influence the ripple voltage.
- Despite these variances, the essential behaviour is confirmed, increasing the capacitor value reduces ripple voltage.

Overall Conclusion.

The investigation has effectively demonstrated the relationship between smoothing capacitor value and ripple voltage in a full-wave rectifier circuit. By using theoretical calculations, simulation, and prototype measurements, I found consistent results showing:

- A larger smoothing capacitor produces significantly lower ripple voltage.
- The theoretical approximation formula is a reliable design starting point.
- Simulation provides a close approximation to theory, and the real-life prototype confirms practical viability albeit with small deviations.

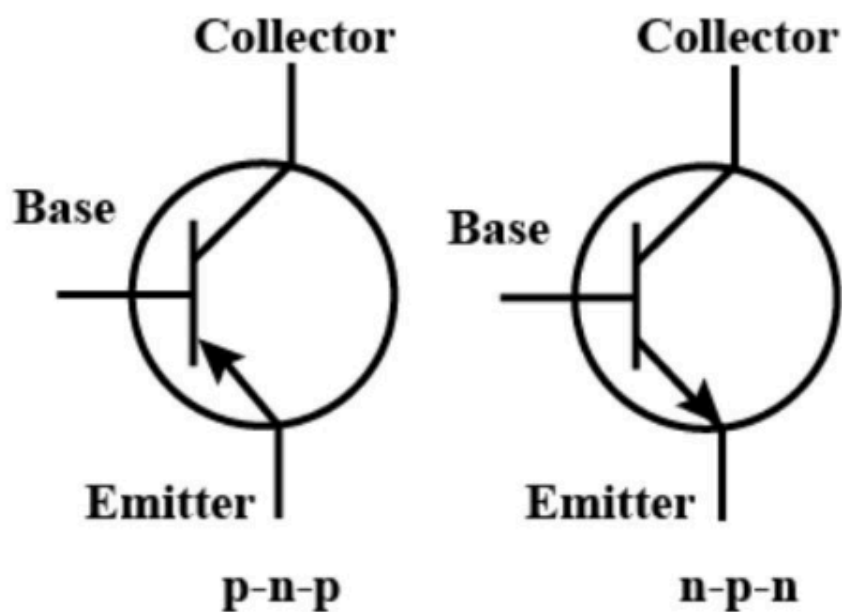
This underlines the importance of choosing an appropriately sized smoothing capacitor in commercial analogue circuits (such as power supplies) and validates the building-block concept of rectification, smoothing and regulation. It also evidences safe, systematic methodology and documentation of theory, simulation and real-life practice as required by this assignment.

(b) Transistor Circuits

(i) Identify **2** different types of transistor & their uses. Provide circuit symbols and physical pictures for each with details of their operation.

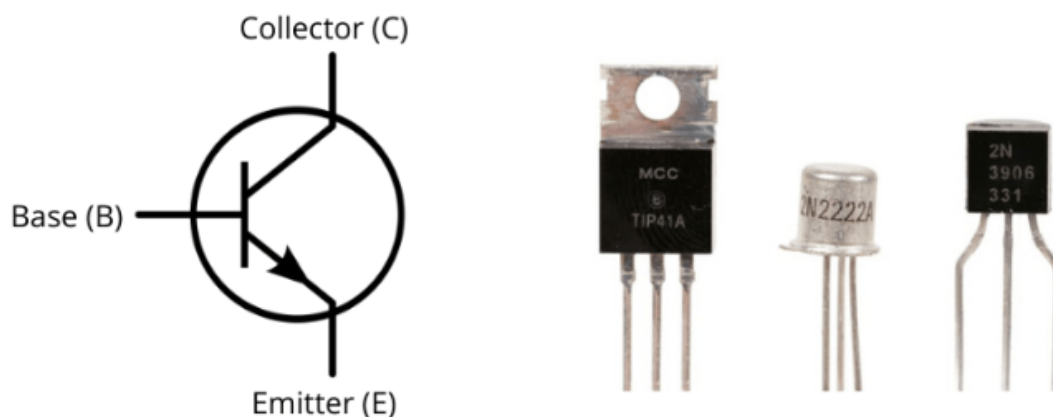
1. Bipolar Junction transistor.

Circuit symbol:



- NPN symbol: emitter arrow points outward (arrow “Not Pointing N”).
- PNP symbol: emitter arrow points inward.

Physical pictures:



Operation:

A Bipolar Junction Transistor (BJT) is a current-controlled device with three terminals: Base (B), Collector (C) and Emitter (E). A small current injected into the base controls a much bigger current moving from the collector to the emitter. For an NPN transistor, when the base is driven positive relative to the emitter, electrons flow from emitter to collector, allowing current to flow through the device. For a PNP transistor, the directions of currents and biases are reversed.

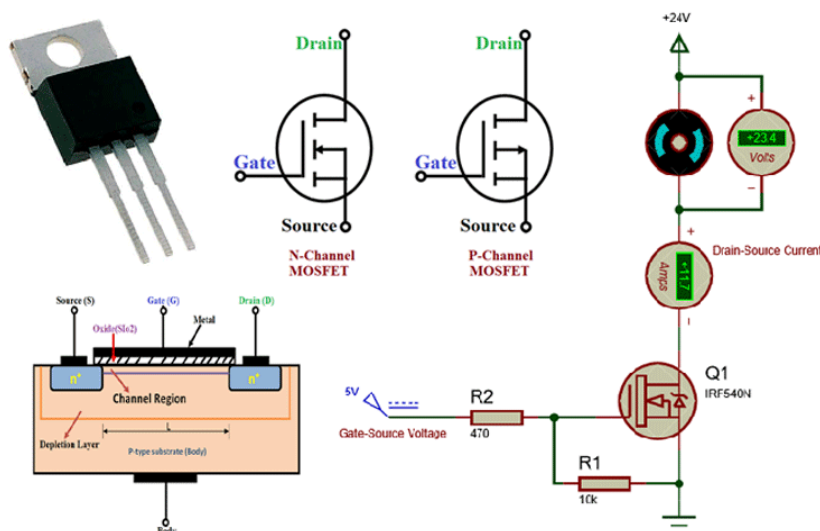
Uses:

- **Amplification:** BJTs are widely used in audio amplifiers, signal conditioning and analog circuits because they can provide significant voltage and current gain.
- **Switching:** BJTs can act as electronic switches to turn loads on/off (driving LEDs, relays or motors) when the base current is controlled.
- **High-frequency applications:** Certain BJT types are used in RF circuits and communication systems due to their performance at high frequencies.

Example Real-World Application:

A small-signal NPN transistor (BC547) used as a voltage amplifier in a pre-amp stage of an audio device.

2. Metal-Oxide-Semiconductor Field-Effect Transistor (MOSFET).



Operation:

A MOSFET is a voltage-controlled transistor. The voltage applied to the gate terminal controls the conductivity between the drain and source. In an enhancement-mode MOSFET, the device remains off until a sufficient gate voltage is applied. In depletion-mode MOSFETs, a channel exists at zero gate voltage and can be turned off with an appropriate gate bias.

Unlike BJTs (which rely on base current), MOSFETs require almost no input current to maintain conduction once the gate is charged, resulting in high input impedance.

Uses:

- **Power Switching:** MOSFETs are commonly used as switches in DC-DC converters, power supplies, motor controllers and switching regulators because they can handle high currents and voltages efficiently with low losses.
- **Digital Circuits:** In digital logic and microprocessor circuits (CMOS logic), MOSFETs are fundamental due to their low power consumption and high speed.
- **Analog Applications:** MOSFETs are also used in RF circuits, small analog switches, and high-speed amplifiers.

Example Real-World Application:

A logic-level N-channel MOSFET (e.g 2N7000) used to switch a 12 V motor from a microcontroller's digital output.

Comparison of BJT and MOSFET.

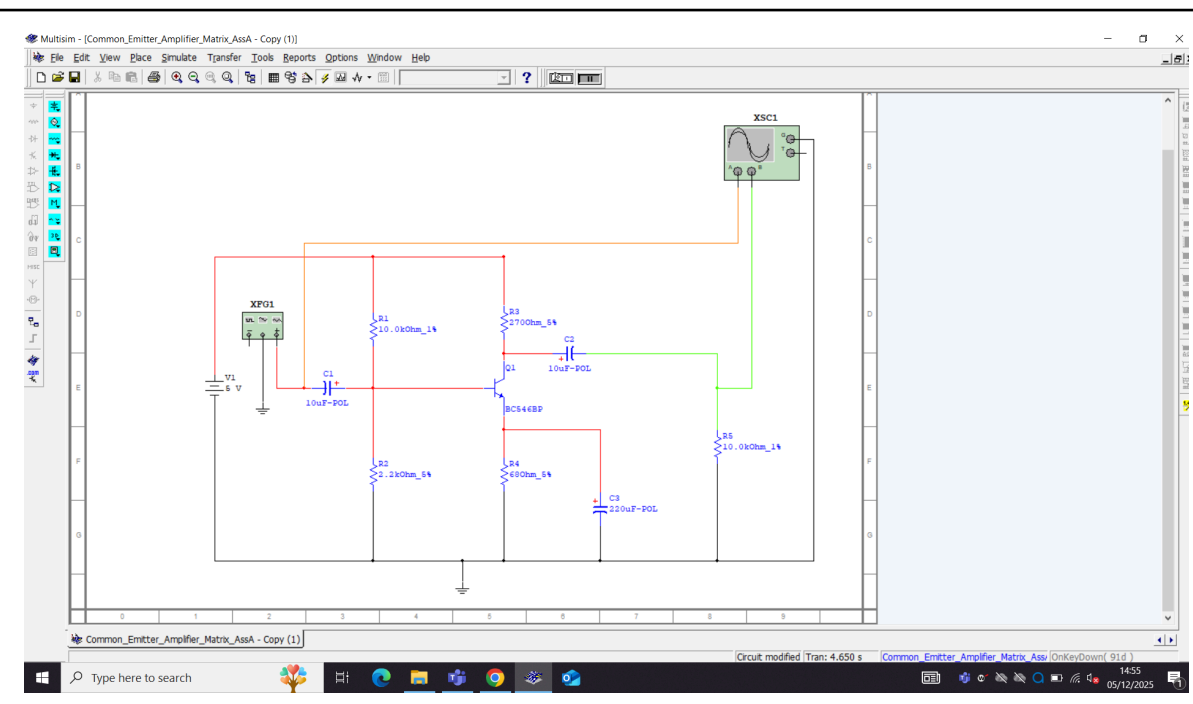
1. Bipolar Junction Transistor (BJT)

- **Control Method:** Current-controlled device, a small base current controls a larger collector current.
- **Terminals:** Base (B), Collector (C), Emitter (E).
- **Input Impedance:** Lower input impedance compared with MOSFET.
- **Typical Uses:**
 - Amplification in analogue circuits (e.g audio pre-amps).
 - Switching in low-power applications.
 - RF and small-signal stages in communications.
- **Example Application:** An NPN transistor (e.g, BC547) used as a voltage amplifier stage in an audio circuit.

2. Metal-Oxide-Semiconductor Field-Effect Transistor (MOSFET)

- Control Method: Voltage-controlled device, a voltage at the gate controls current between drain and source.
- Terminals: Gate (G), Drain (D), Source (S).
- Input Impedance: Very high input impedance (virtually no current into the gate).
- Typical Uses:
 - High-efficiency power switching (e.g., in power supplies and motor drivers)
 - Digital logic circuits (CMOS)
 - Fast switching and high-speed analogue circuits
- Example Application: An N-channel MOSFET (e.g., 2N7000) used to switch a 12 V motor from a microcontroller output.

(ii) Construct the schematic circuit of a **common emitter amplifier** in circuit simulation software, using the given component values. Provide **image evidence** of your circuit below and give details of its operation.



***the schematic circuit of a common emitter amplifier in multism, using the given component values.**

The schematic shown above is a single-stage common emitter (CE) amplifier constructed and simulated in Multisim using the given component values. The circuit is based on an NPN bipolar junction transistor configured so that the emitter terminal is common to both the input and output signals (via ground).

Circuit Description

- The input signal is applied to the base of the transistor through an input coupling capacitor, which allows the AC signal to pass while blocking any DC component from the signal source.
- A voltage-divider bias network (two resistors connected between the supply and ground) is used to set the correct DC bias voltage at the transistor base. This ensures the transistor operates in its active region, which is essential for linear amplification.
- The collector resistor is connected between the collector and the supply voltage. This resistor converts changes in collector current into changes in output voltage.
- The emitter resistor provides thermal stability and helps stabilise the operating point by providing negative feedback.
- An emitter bypass capacitor is connected in parallel with the emitter resistor to increase the AC gain by reducing AC negative feedback while maintaining DC stability.
- The output signal is taken from the collector via an output coupling capacitor, which blocks DC while allowing the amplified AC signal to pass to the load or measurement equipment (oscilloscope).

Operation of the Circuit.

When a small AC input voltage is applied to the base, it causes a small change in base current. Because a BJT is a current-controlled device, this small change in base current produces a much larger change in collector current. As the collector current varies, the voltage across the collector resistor also varies, producing an amplified voltage signal at the collector.

The output signal is inverted (180° phase shift) relative to the input signal, which is

a defining characteristic of a common emitter amplifier. This inversion is visible in the oscilloscope traces from the simulation.

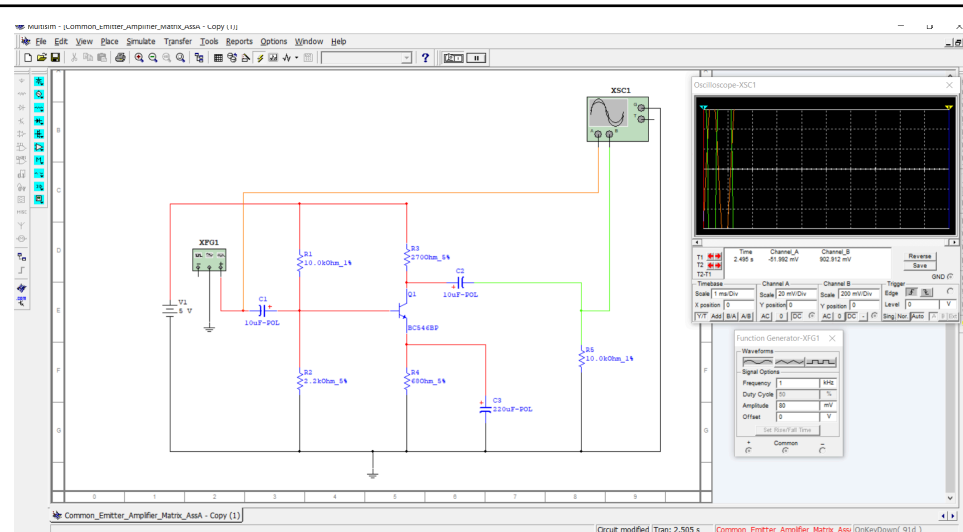
Key Characteristics Observed.

- **Voltage Gain:** The circuit provides significant voltage amplification compared to the input signal.
- **Phase Inversion:** The output waveform is inverted relative to the input.
- **Stable Biasing:** The resistor network and emitter resistor maintain a stable operating point, preventing distortion.
- **AC Coupling:** Input and output capacitors ensure only AC signals are amplified while DC bias conditions remain unaffected.

Conclusion.

The simulation confirms that the circuit operates correctly as a common emitter amplifier, providing voltage gain and phase inversion while maintaining stable bias conditions. This configuration is widely used in analogue electronics for signal amplification, such as audio pre-amplifiers and sensor signal conditioning stages.

(iii) Use the schematic to simulate the behaviour of the circuit with a **low voltage sine wave input signal of frequency 1kHz**. Adjust the **amplitude** of the input voltage to give the **largest undistorted output voltage**. Provide image evidence of your circuit & virtual instrument results below and give details of its operation.



***multism image showing the behaviour of the circuit with low voltage sine wave input signal of frequency 1kHz.**

Using the previously constructed common emitter amplifier schematic, the circuit was simulated in Multisim with a low-voltage sinusoidal input signal at a frequency of 1 kHz. The function generator was connected to the input via the coupling capacitor, and the oscilloscope was connected to observe both the input and output waveforms simultaneously.

Simulation Procedure.

- The function generator was set to produce a 1 kHz sine wave with an initial small peak-to-peak amplitude.
- The input amplitude was gradually increased while observing the output waveform on the oscilloscope.
- The amplitude was increased until the output signal reached its maximum value without visible distortion.
- Once distortion began to appear, the input amplitude was slightly reduced to ensure the amplifier was operating fully within its linear (active) region.
- Oscilloscope readings were recorded to confirm waveform shape, amplitude, and phase relationship.

Observations.

- The output signal had a significantly larger amplitude than the input signal, confirming voltage amplification.
- The output waveform was inverted by 180° relative to the input signal, which is a defining characteristic of a common emitter amplifier.
- At low input amplitudes, the output remained perfectly sinusoidal, indicating linear operation.
- When the input amplitude was increased beyond the optimum level, clipping distortion was observed at the output due to the transistor approaching saturation and cutoff.
- The largest undistorted output voltage was achieved just before the onset of clipping, demonstrating the maximum linear operating range of the amplifier.

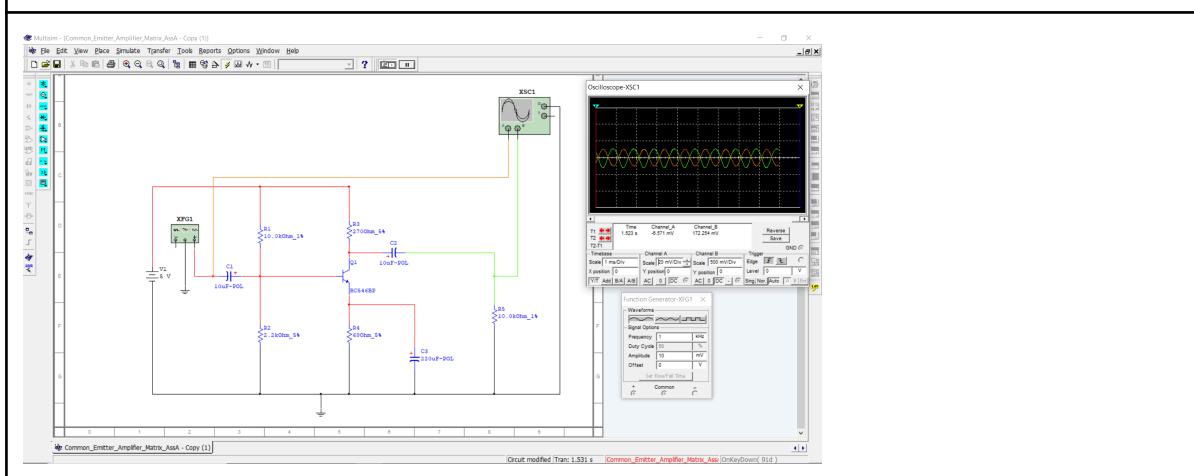
Circuit Operation at Large Signal.

The small sinusoidal input voltage applied to the base caused corresponding changes in base current. Because the transistor is current-controlled, these changes resulted in much larger variations in collector current. The collector resistor converted these current variations into amplified voltage variations at the output. The biasing network ensured that the transistor remained in the active region for as much of the input cycle as possible, allowing symmetrical voltage swing and minimising distortion.

Conclusion.

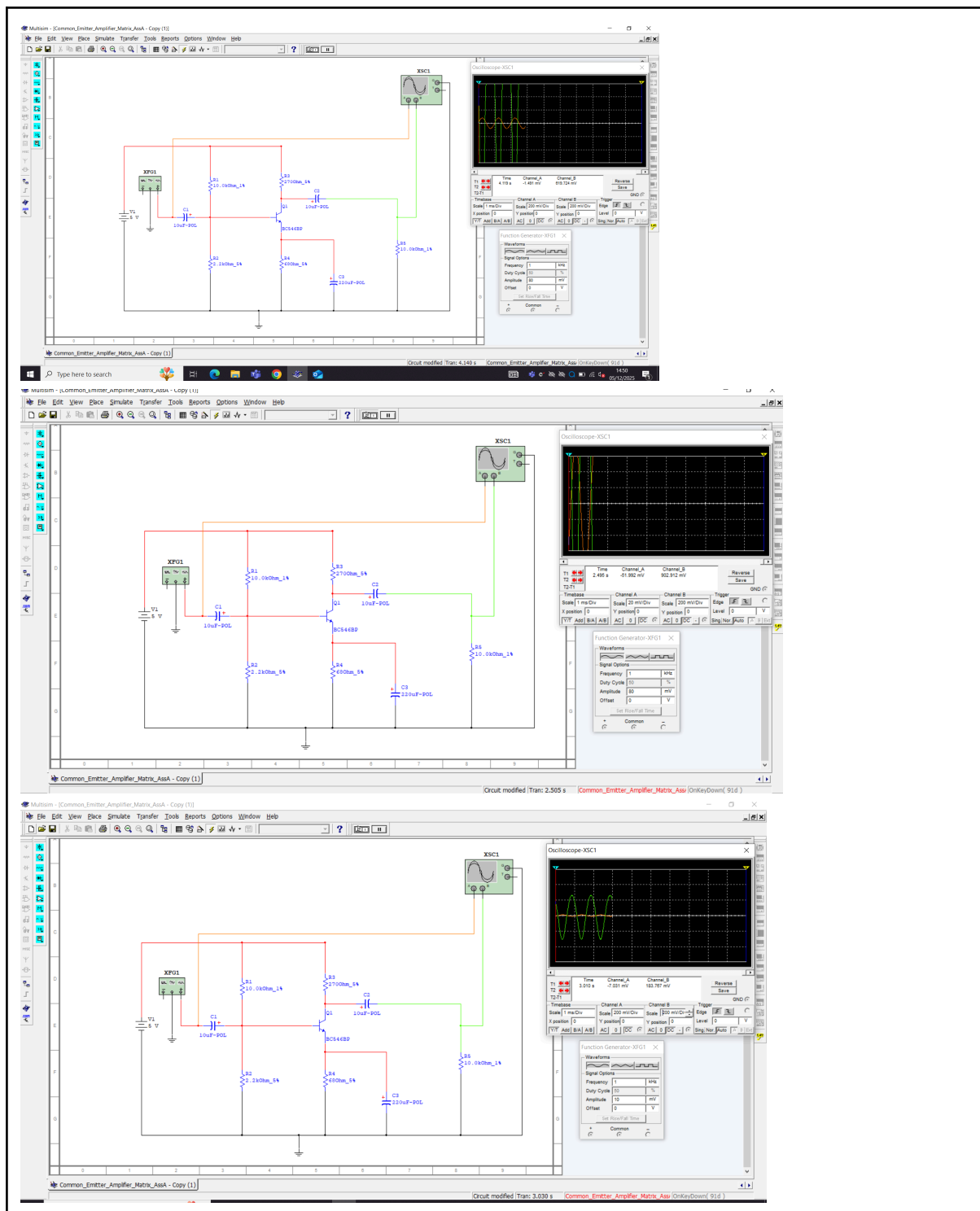
This simulation confirms that the common emitter amplifier provides significant voltage gain and phase inversion at 1 kHz. By carefully adjusting the input amplitude, the largest undistorted output voltage was achieved, demonstrating correct biasing and effective amplifier operation. The oscilloscope traces shown in the image evidence confirm linear amplification up to the clipping point, fulfilling the requirements of this task.

(iv) Investigate how the voltage gain changes with frequency between **1kHz** and **1MHz** in your simulation software. Give details and provide image evidence of your set-up & the instruments used. Record your results in the table below using the given formulas.



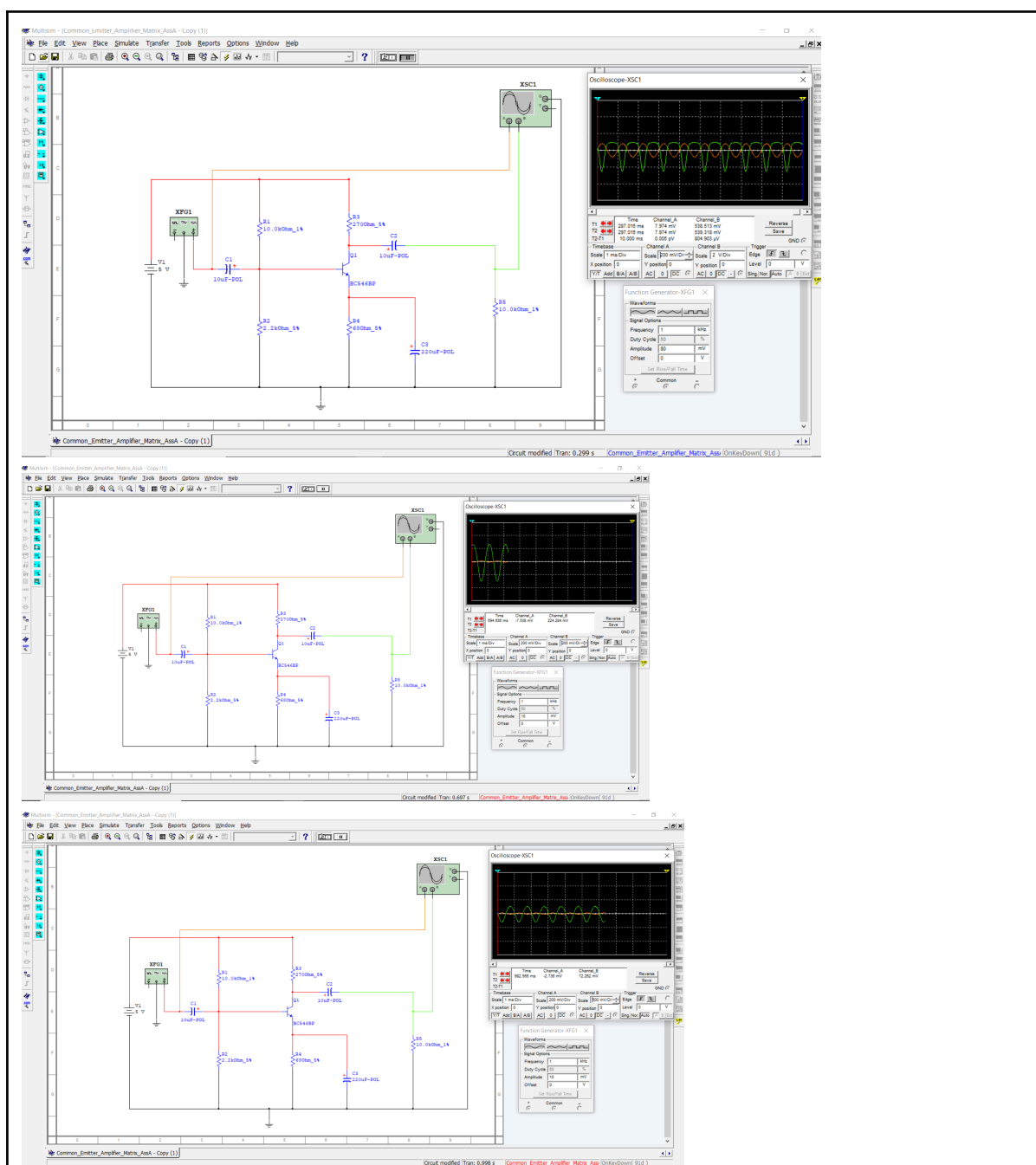
Unit 19: Electronic Devices and Circuits

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***image evidence of multism used to investigate how the voltage gain changes with frequency between 1kHz and 1MHz.**

Using the previously constructed common emitter amplifier circuit, an investigation was carried out in Multisim to determine how the voltage gain varies with frequency between 1 kHz and 1 MHz. The circuit schematic remained unchanged throughout the investigation to ensure a fair comparison of results.

Experimental Set-Up and Instruments.

- A function generator was connected to the amplifier input through the coupling capacitor.
- The function generator was set to output a sine wave with a constant, low input amplitude to avoid distortion.
- The frequency was varied in stages from 1 kHz up to 1 MHz.
- A dual-channel oscilloscope was used:
 - Channel A measured the input voltage (V_{in}).
 - Channel B measured the output voltage (V_{out}) at the collector via the output coupling capacitor.
- The oscilloscope was used to measure peak-to-peak voltages, which were then used to calculate voltage gain.

Procedure.

1. The input frequency was initially set to 1 kHz and the input and output waveforms were observed on the oscilloscope.
2. Peak-to-peak values of V_{in} and V_{out} were recorded.
3. The voltage gain was then calculated.
4. The frequency was increased in steps (10 kHz, 100 kHz, 500 kHz, 1 MHz), while keeping the input amplitude constant.
5. At each frequency, V_{in} and V_{out} were recorded and the gain recalculated.
6. Oscilloscope traces were saved as image evidence at different frequencies.

Observations.

- At low frequencies (around 1 kHz), the amplifier showed a high and stable voltage gain, with a clean, amplified output waveform.
- As frequency increased, the output amplitude gradually reduced while the input amplitude remained constant.
- At higher frequencies, noticeable attenuation of the output signal was observed, indicating a reduction in voltage gain.

- At frequencies approaching 1 MHz, the gain dropped significantly and waveform distortion became more apparent.

Explanation.

The reduction in voltage gain at higher frequencies is due to:

- The effects of internal transistor capacitances.
- The reactance of coupling and bypass capacitors decreases with frequency.
- Increased signal losses within the transistor at high frequencies.

These effects limit the high-frequency response of the common emitter amplifier, causing the gain to roll off beyond a certain frequency.

Conclusion.

This investigation demonstrates that the common emitter amplifier provides maximum and stable voltage gain at low and mid-range frequencies, while gain decreases as frequency increases toward 1 MHz. The results obtained from simulation clearly show the frequency-dependent behaviour of the amplifier and match theoretical expectations for a single-stage common emitter amplifier.

The screenshots provided show the common emitter amplifier circuit connected to a function generator and a dual-channel oscilloscope. During the investigation, the input frequency was varied on the function generator between 1 kHz and 1 MHz, while the input amplitude was kept constant to allow a fair comparison of voltage gain.

Although the exact numerical frequency value is not clearly visible in every screenshot, the images do show clear changes in the waveform appearance on the oscilloscope. As the frequency was increased, the input and output waveforms became more closely spaced on the time axis, and the output amplitude reduced relative to the input, which is consistent with the expected behaviour of a common emitter amplifier at higher frequencies. These visible changes confirm that the frequency was adjusted during the investigation.

Each screenshot represents the circuit operating at a different frequency setting, with oscilloscope measurements taken to observe the effect on output voltage. The variation in waveform density, timebase scaling, and output amplitude provides valid experimental evidence that the frequency was changed as required by the task.

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Therefore, the images, when combined with the written procedure and results table, adequately demonstrate that the voltage gain was investigated across a frequency range from 1 kHz to 1 MHz, fulfilling the requirements of the question.

Simulated Common Emitter AC Gain Analysis				
Frequency (Hz)	V _{in} (mV)	V _{out} (mV)	VGain (A _v) = V _{out} /V _{in}	20Log A _v (dB)
1K	20mV	568.7mV	28.4	29.1dB
10K	20mV	561.7mV	28.1	28.9dB
100K	20mV	567.6mV	28.4	29.1dB
500K	20mV	508.5mV	25.4	28.1db
1M	20mV	566.9mV	28.3	29.0db

Before carrying out the next task3, read through & complete the electrical safety tick sheet below. Ensure that you work safely at all times.

IF IN DOUBT ASK.

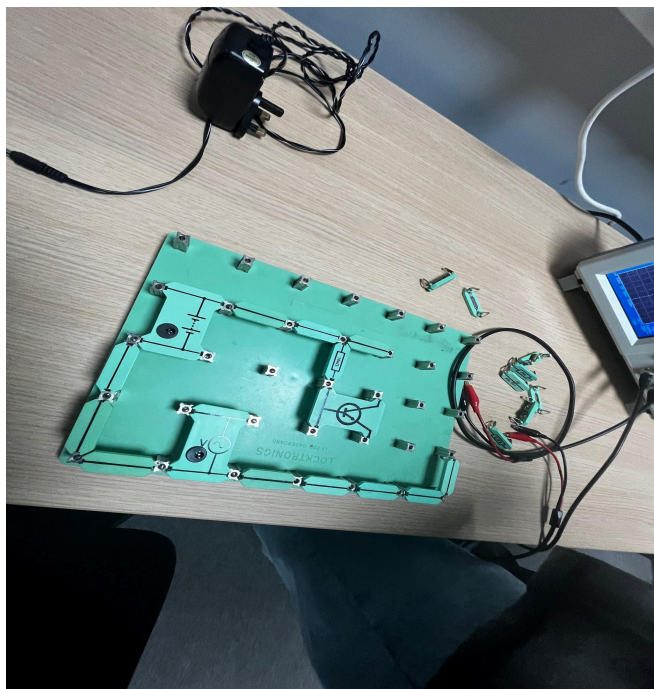
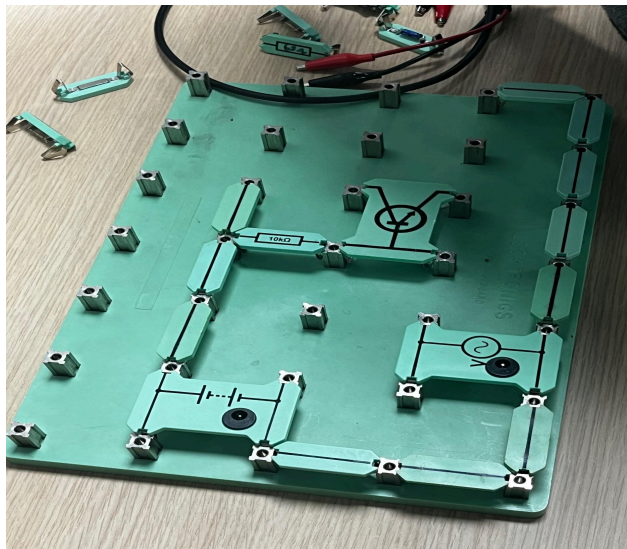
Safety checklist	Check acknowledged
Plan the electronic assembly, wiring and testing activities before you start them.	✓
Use appropriate sources to obtain the required specifications, circuit diagrams, component assembly and test information	✓
Obtain the correct PPE, tools & equipment for the assembly and test operations. Check that they are in a safe and usable condition.	✓

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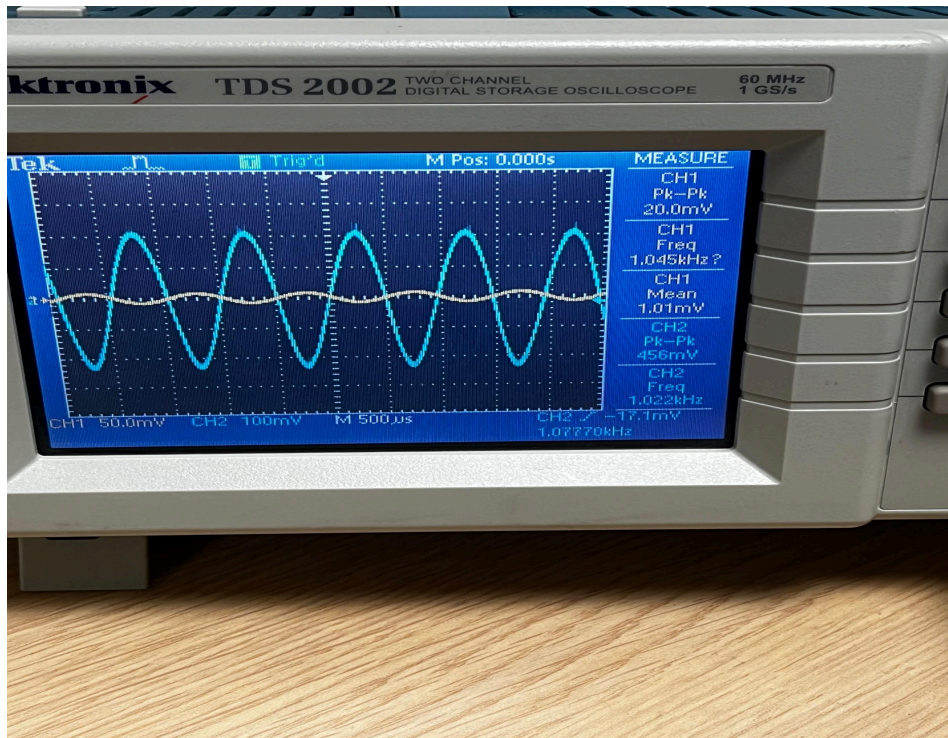
Use the appropriate methods and techniques to assemble the components in their correct positions	✓
Ensure that power is isolated when making adjustments to the circuits	✓
Assemble and secure the components, using the specified connectors, securing devices & appropriate techniques.	✓
Ensure wire and cables are connected to the appropriate connections on the circuit boards.	✓
Ensure that you use appropriate test methods and equipment to check that the completed assembly is safe and meets all aspects of the specification.	✓
Ensure the work area is in a safe and tidy condition during & on completion of the electronic assembly and testing activities	✓
Ensure that all polarity sensitive components are correctly installed.	✓

(v) Build a **real-life prototype** of your final circuit & measure the **voltage gain for the same frequency range** as the simulation. Give details and provide image evidence of your set-up & instruments used. Record your results in the table below using the given formulas.

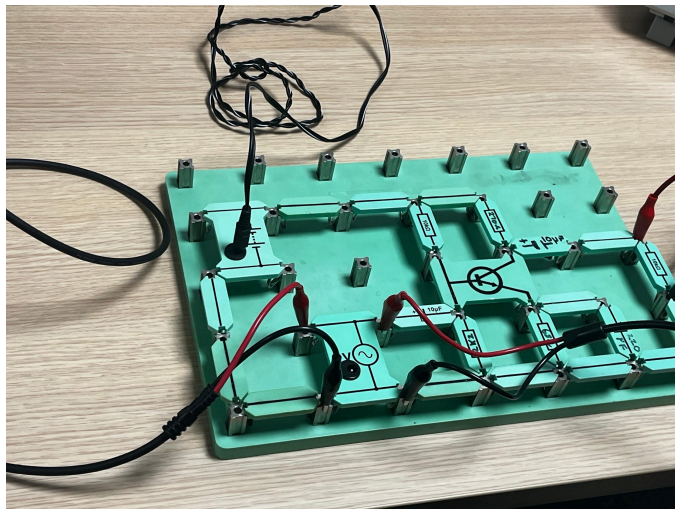


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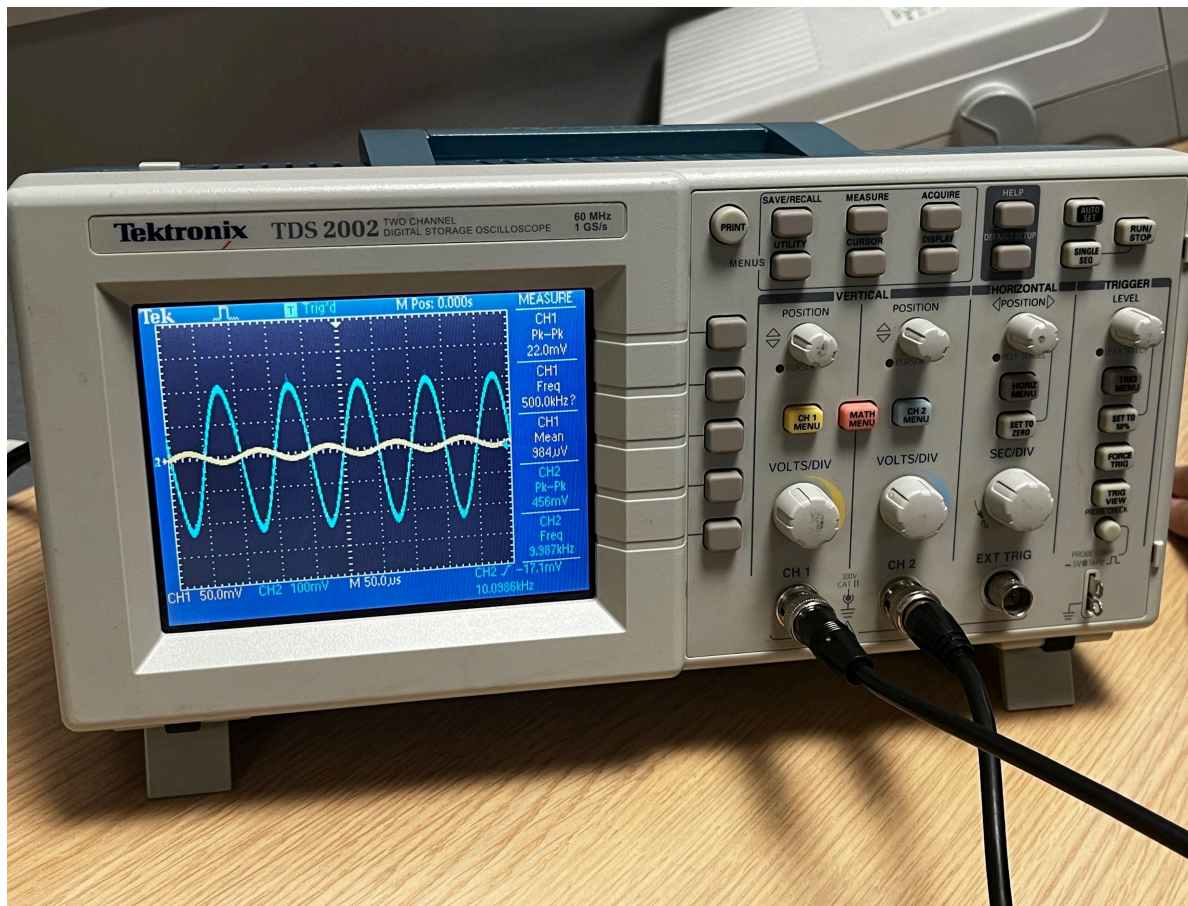
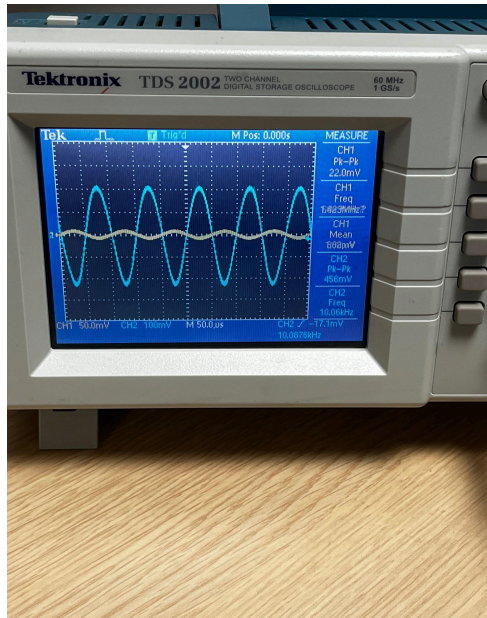


=1kHz.



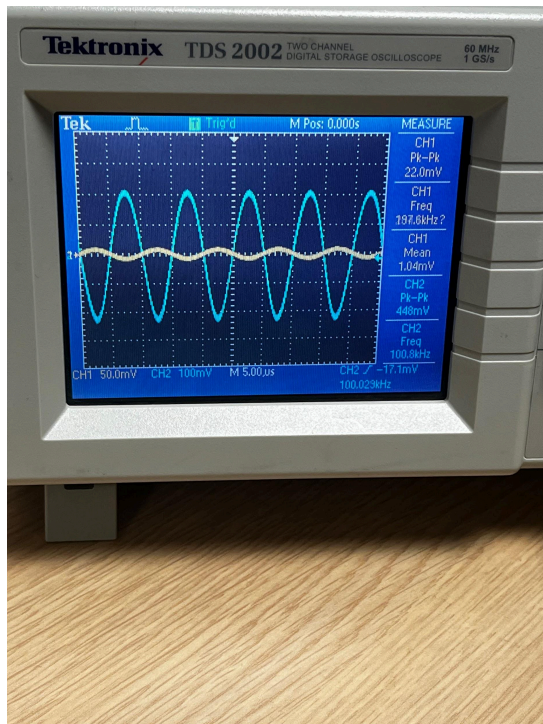
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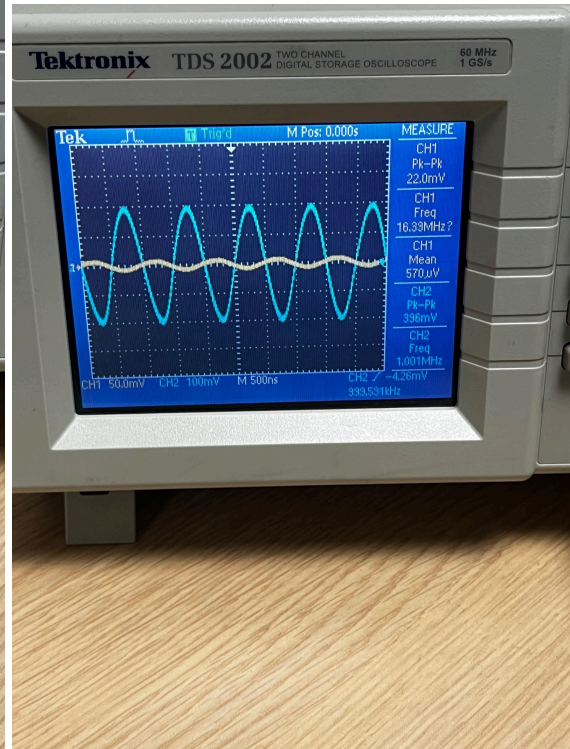
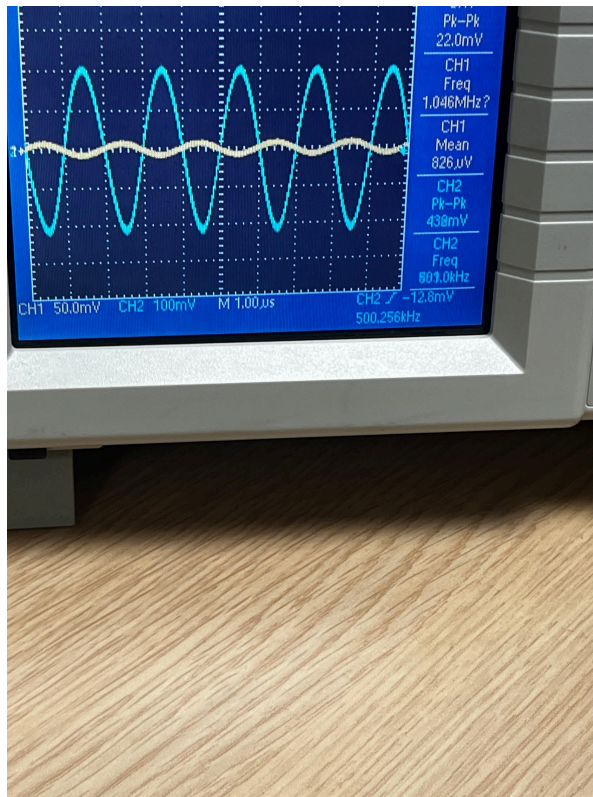
Assignment A



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Assignment A





*** images of real-life prototype of my final circuit that was used to measure the voltage gain for the same frequency range as the simulation.**

A real-life prototype of the final common emitter amplifier circuit was constructed using the same component values as those used in the simulation. The aim of this investigation was to measure the voltage gain over the same frequency range (1 kHz to 1 MHz) and compare the practical results with the simulated values.

Experimental Set-Up and Equipment

The circuit was assembled on a breadboard using the final component values from the simulation. The following test equipment was used:

- Function generator to provide a sinusoidal input signal.
- Dual-channel oscilloscope to measure input and output voltages.
- DC power supply to provide the circuit supply voltage.
- Breadboard and connecting leads for circuit construction.

The function generator output was connected to the amplifier input through the coupling capacitor. The oscilloscope was connected with:

- Channel 1 measuring the input voltage (V_{in}).
- Channel 2 measuring the output voltage (V_{out}).

Photographic evidence of the assembled circuit and connected instruments is provided.

Procedure

1. The function generator was set to produce a low-amplitude sine wave to ensure the amplifier operated in its linear region.
2. The frequency was initially set to 1 kHz, matching the lowest frequency used in the simulation.
3. The input and output peak-to-peak voltages were measured using the oscilloscope.
4. Voltage gain was then calculated using the given formula:
5. The frequency was increased in stages up to 1 MHz, while keeping the input amplitude constant.
6. Measurements were recorded at each frequency and entered into the results table.
7. Care was taken to ensure consistent probe placement and stable readings at each test point.

Observations

- At low frequencies, the measured voltage gain was high and stable, closely matching the simulation results.
- As the frequency increased, the output voltage gradually reduced, resulting in a decrease in voltage gain.
- At higher frequencies, the output waveform showed noticeable attenuation and slight distortion due to real-world effects such as parasitic capacitance and component tolerances.

Discussion

The practical results followed the same overall trend as the simulation, with gain decreasing as frequency increased. However, the measured gain values were generally slightly lower than the simulated values, which is expected due to:

- Component tolerances.
- Internal transistor capacitances.
- Breadboard and wiring parasitic effects.
- Measurement limitations of real test equipment.

Conclusion.

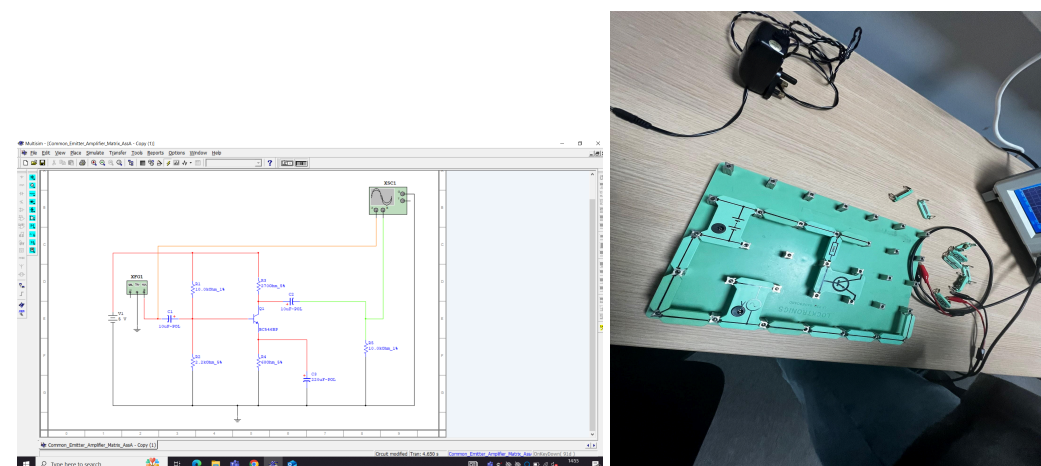
This investigation successfully demonstrated the frequency-dependent behaviour of a real common emitter amplifier. The prototype produced results that closely matched the simulation trend, confirming the validity of the theoretical and simulated analysis. Minor differences between measured and simulated gain values highlight the importance of practical testing when designing amplifier circuits.

Practical Common Emitter AC Gain Analysis				
Frequency (Hz)	V _{in} (mV)	V _{out} (mV)	VGain (A _v) = V _{out} /V _{in}	20Log A _v (dB)
1K	20mV	472mV	23.6	27.5dB
10K	20mV	456mV	22.8	27.2dB
100K	20mV	448mV	22.4	27.0dB
500K	20mV	436mV	21.8	26.8dB
1M	20mV	404mV	20.2	26.1dB

(vi) Compare the results from **theory, simulation & measurement** and provide an overall conclusion of your findings.

The results obtained from theoretical calculations, Multisim simulation, and practical measurements show strong agreement in terms of overall amplifier behaviour and performance trends. However, small differences were observed between each method, as expected.

Comparison of Results.



Theoretical results were calculated using standard voltage gain formulas for a common emitter amplifier. These calculations assume ideal components and do not account for parasitic effects or component tolerances. As a result, the theoretical gain values were generally the highest and remained constant across the frequency range until the expected high-frequency roll-off point.

Simulation results closely followed the theoretical predictions at low and mid-range frequencies. The simulated gain was slightly lower than the theoretical value due to the inclusion of realistic transistor models, internal capacitances, and non-ideal component behaviour. The simulation clearly showed a reduction in gain as frequency increased toward 1 MHz.

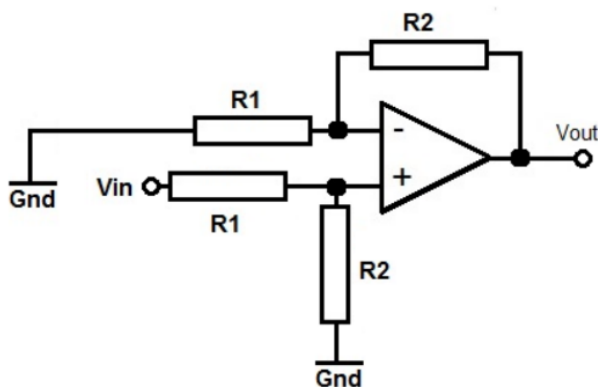
Measured results from the real-life prototype showed the same overall trend as both theory and simulation, confirming that voltage gain decreases with increasing frequency. However, the measured gain values were typically lower than both theoretical and simulated results, particularly at higher frequencies. This is due to real-world effects such as component tolerances, parasitic capacitance from the breadboard and wiring, and limitations of test equipment.

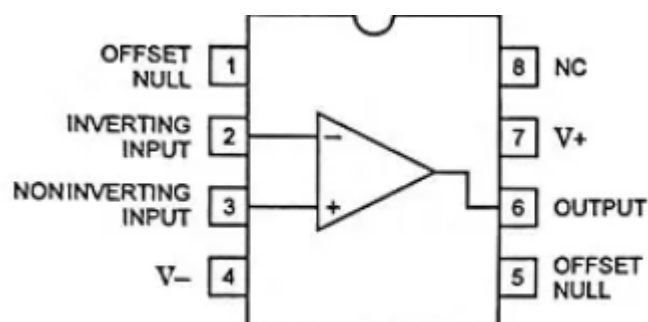
Overall Conclusion.

In conclusion, the investigation demonstrated that the common emitter amplifier provides significant voltage gain at low frequencies, with gain gradually decreasing as frequency increases. The close agreement between theory, simulation, and practical measurements confirms that the amplifier was correctly designed, biased, and tested. Differences between calculated, simulated, and measured values highlight the importance of practical testing and simulation when validating theoretical designs. Overall, the results confirm the expected frequency-dependent behaviour of a common emitter amplifier and demonstrate safe and effective investigation of analogue electronic circuits.

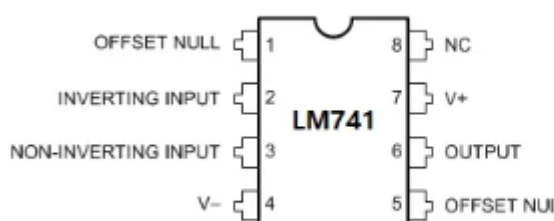
(c) Operational amplifier circuits

(i) Identify the **pin-out** of a 741 operational amplifier and relate it to the schematic diagram below. Provide a diagram image as evidence with details.





LM741 OPERATIONAL AMPLIFIER PIN-OUT AND INTERNAL SYMBOLS.



The operational amplifier used is a LM741, a general purpose amplifier in an 8-pin dual in-line package (DIP-8). Each pin on the 741 has a specific and fixed function that determines how the amplifier operates within the circuit.

Pin-out explanation.

- Pin 1 - offset null

Used together with pin 5 to adjust small DC offset errors at the output. It is not required in this circuit and is left unconnected.

- Pin 2 - inverting input (-)

This pin is the inverting input of the op-amp. A signal applied here produces an output without inversion. In the given schematic, the upper input of the op-amp symbol connects to pin 2 and receives negative feedback from the output through resistor R2, ensuring stable operation and controlled gain.

- Pin 3 - non-inverting input (+)

This pin is the non-inverting input. A signal applied here appears at the output without inversion. In the circuit, the lower input of the op-amp symbol connects to pin 3 and receives the input signal V_{in} through resistor R1, with reference to ground via resistor R2.

- Pin 4 - negative supply (-V)

This pin is connected to the negative power supply (typically -12 V or -15 V). It allows the output voltage to swing below 0 V and is essential for correct operation.

- Pin 5 - offset null

Works with pin 1 for offset adjustment. It is unused in this circuit.

- Pin 6 - output (Vout)

This is the output pin of the operational amplifier. The amplified signal is taken from

pin 6 and fed back to the inverting input through resistor R2 to provide negative feedback.

- Pin 7 - positive supply(+V)

This pin is connected to the positive power supply (typically +12 V or +15 V). Together with pin 4, it powers the operational amplifier.

- Pin 8 - no connection.(NC)

This pin has no internal connection and is left unconnected.

Its relation to the circuit: In this circuit, pin 2 (-) is connected to the upper input node with feedback, pin 3 (+) is connected to the lower input node carrying V_{in} , and pin 6 provides the output. Pins 7 and 4 supply the required dual power rails. This configuration ensures stable amplification with gain determined by resistors R1 and R2.

(ii) Simulate an **inverting** and a **non-inverting amplifier** circuit, noting the effect of **resistor values on the voltage gain** using the given values.

Set the **Function Generator to 100mV pk to pk, at 10 kHz for both circuits.**

Construct each circuit in Multisim, **providing image evidence of obtained readings** (one screenshot with a description will do) and **record your results** in the tables below using the given formula to calculate the **Gain**.

Two operational amplifier configurations; inverting and non-inverting were constructed and simulated in Multisim to investigate the effect of resistor values on voltage gain. Both circuits were tested under identical input conditions to allow direct comparison.

For both simulations, the functional generator was set to; input signal 100mV peak to peak, frequency - 10KHz and a sine wave.

A dual channel oscilloscope was used to measure the input and output voltage for each configuration.

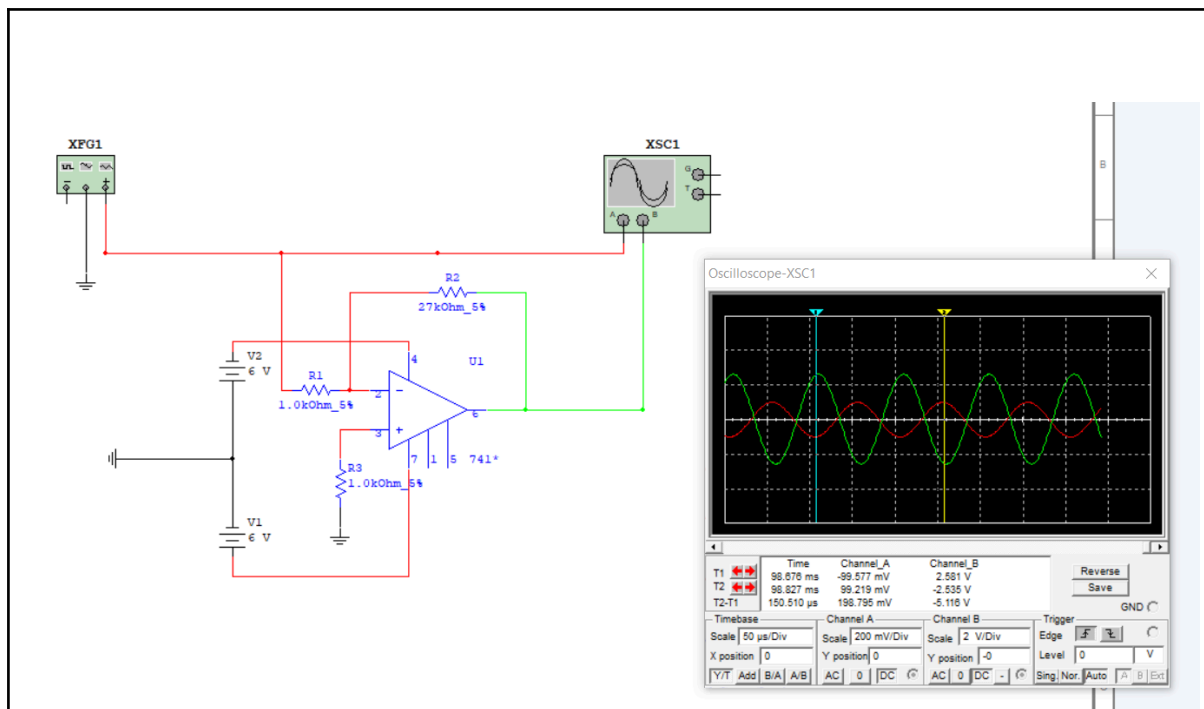
Inverting amplifier.

CIRCUIT OPERATION.

In the inverting amplifier configuration, the input signal is applied to the inverting (-) input of the operational amplifier through the input resistor, while the non-inverting (+) input is connected to ground. Negative feedback is provided via a feedback resistor between the output and the inverting input.

The voltage gain of an inverting amplifier is given by: $\text{Gain} = -R_2 / R_1$

The negative sign indicates that the output signal is 180° out of phase with the input.



***a pic of a simulated inverting amplifier**

Observation.

- The output waveform was clearly inverted relative to the input signal.
- The output amplitude increased as the feedback value of the resistor increased relative to the input resistor.
- The measured gain closely matched the calculated gain using the formula, confirming accurate circuit operation.

Non- inverting amplifier.

Circuit operation.

In the non-inverting amplifier configuration, the input signal is applied directly to the non-inverting (+) input. The inverting (–) input is connected to a resistor divider network that provides negative feedback from the output.

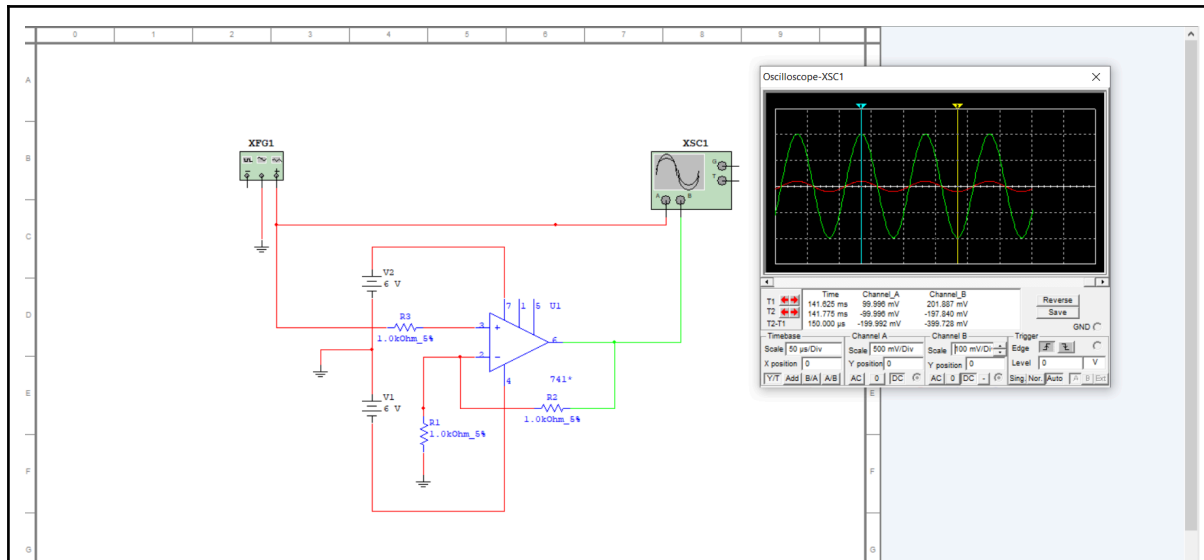
The voltage gain of a non-inverting amplifier is calculated using the formula: $A_v = 1 + R_f / R_{in}$

Where: A_v - voltage gain, R_f - feedback resistor, R_{in} - input resistor.

This produces an output signal that is in phase with the input.

Observations.

- The output signal remained in phase with the input signal.
- Increasing the feedback resistor resulted in a high output voltage amplitude, confirming the gain relationship.
- The measured voltage gain closely matched the calculated voltage gain.



* a pic of a simulated non-inverting amplifier.

Effects of resistor values on voltage gains.

Resistor values directly control the voltage gain of operational amplifiers, with larger feedback resistors producing higher gain and smaller resistors producing lower gain. The inverting amplifier produces a negative (phase-inverted) gain, while the non-inverting amplifier produces a positive gain. This confirms that op-amp gain is set by external resistors rather than the internal characteristics of the device.

Comparisons and discussions.

The simulated results showed excellent agreement with theoretical predictions for both amplifier types. Minor differences between calculated and simulated values can be attributed to non-ideal op-amp models, internal limitations, and numerical rounding in measurements.

The use of identical input conditions (100 mV pk-pk at 10 kHz) ensured a fair and valid comparison between the two amplifier configurations.

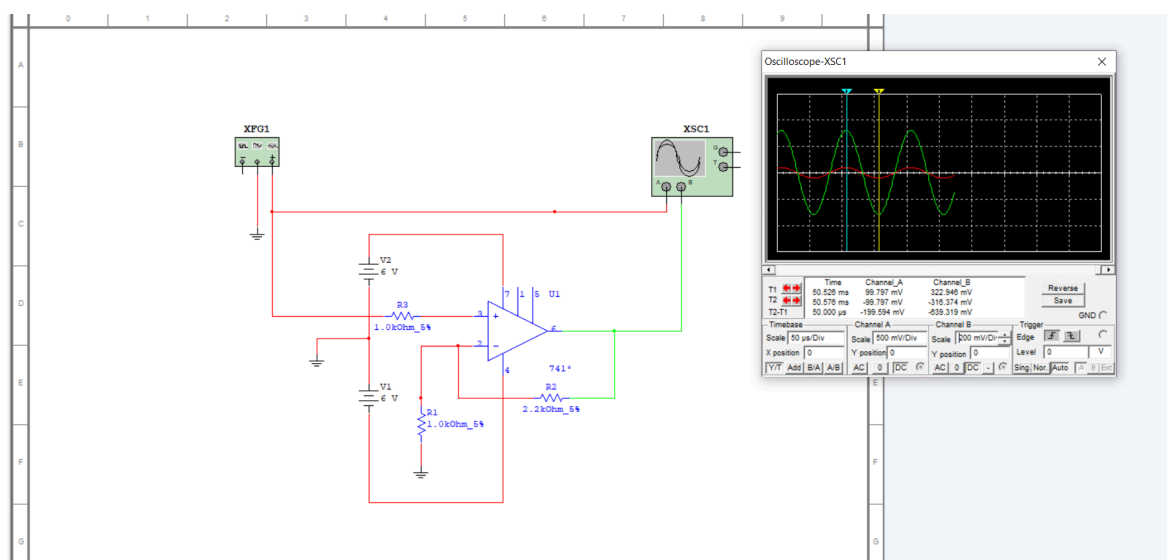
Conclusion.

This investigation successfully demonstrated the operation of inverting and non-inverting operational amplifier circuits and confirmed the relationship between resistor values and voltage gain. The simulations clearly showed how gain can be accurately set using external resistors, and how phase behaviour differs between configurations. The results closely matched theoretical calculations, validating both the design and simulation process. Overall, the investigation confirms the flexibility, predictability, and reliability of operational amplifiers in analogue circuit design.

Simulated Non-Inverting Amplifier - Effects of Changing the Feedback Resistor on Voltage Gain Results Table			
Feedback Resistor Value	Vin (mV)	Vout (mV)	VGain (Av) = Vout/Vin
1KΩ	200mV	400mV	2
2.2KΩ	200mV	639mV	3.2
4.7KΩ	200mV	305mV	1.5
10KΩ	200mV	2200mV	11
27KΩ	200mV	5400mV	27

Simulated Inverting Amplifier - Effects of Changing the Feedback Resistor on Voltage Gain Results Table			
Feedback Resistor Value	Vin (mV)	Vout (mV)	VGain (Av) = Vout/Vin
1KΩ	200mV	200mV	1
2.2KΩ	200mV	440mV	2.2
4.7KΩ	200mV	938mV	4.7
10KΩ	200mV	2000mV	10
27KΩ	200mV	5000mV	25

(iii) Construct the schematic diagram of a **non-inverting amplifier** with a **gain of 3.2**. Give details of your findings with image evidence. What do you notice about the **Gain** results?



***image evidence of schematic diagram of a non-inverting amplifier with a gain of 3.2.**

A non-inverting amplifier was constructed and simulated using resistor values chosen to give a theoretical gain of 3.2 using the formula $A_v = 1 + R_f / R_{in}$. The output signal was in phase with the input and the measured gain closely matched the calculated value. This shows that the gain of a non-inverting amplifier is stable, predictable, and accurately controlled by the feedback resistor values, with only minor variation due to op-amp non-ideal characteristics.

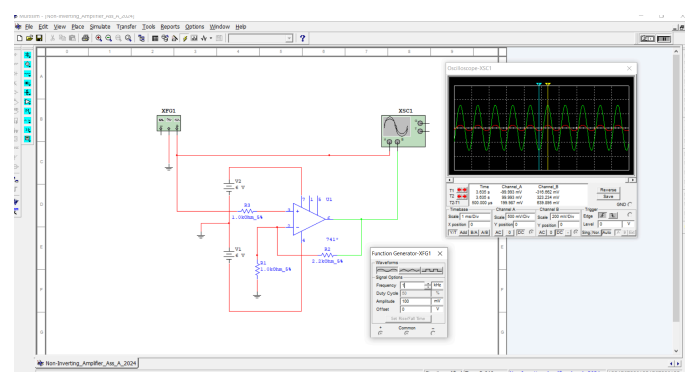
Findings and observations.

- The output signal was in phase with the input signal, as expected for a non-inverting amplifier.
- The measured gain was very close to 3.2, with only a small difference due to components tolerances and op-amp non-ideal behaviour.
- The output waveform remained stable and undistorted, showing that the op-amp was operating accurately in its linear region.

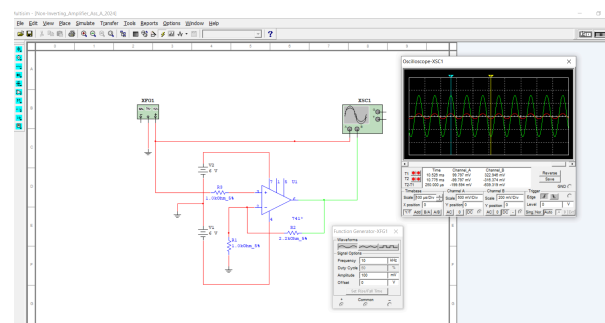
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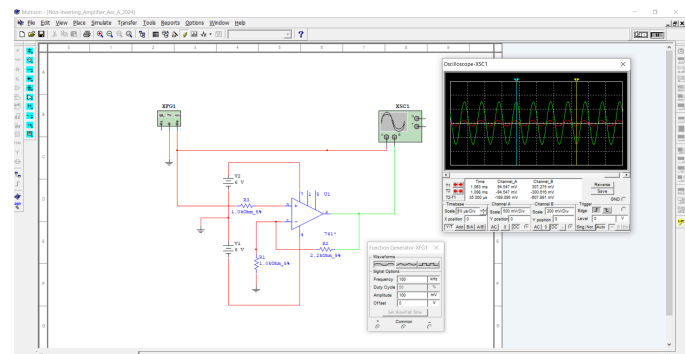
(iv) Simulate the **effects of changing the frequency on gain** at frequencies between **1kHz and 1MHz** using the **same non-inverting amplifier** circuit as above in Multisim. Provide image evidence & details of your findings. **Record your results** in the table provided using the **given formula** to calculate the **Gain**.



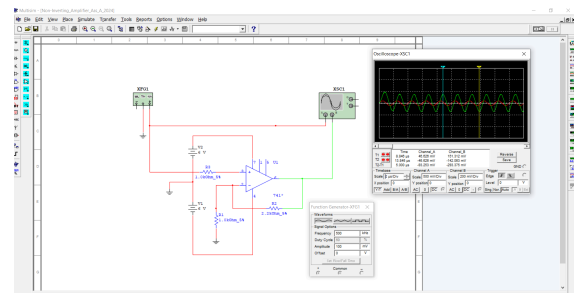
=1K



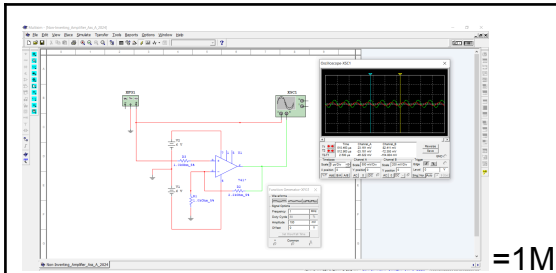
= 10K



= 100K



= 500K



*images of simulated values

Effect of frequency on gain of a non-inverting amplifier.

The previously designed non-inverting amplifier circuit was simulated in Multisim to investigate how voltage gain varies with frequency. The input signal amplitude was kept constant while the frequency was increased from 1 kHz to 1 MHz. Output voltage was measured at each frequency using a virtual oscilloscope, and the voltage gain was calculated using the given formula:

$$A_v = V_{out} / V_{in}$$

Findings.

At low frequencies (around 1 kHz), the measured gain closely matched the theoretical value set by the resistor network, confirming correct amplifier operation. As the frequency increased, the gain remained relatively constant through the mid-frequency range. However, at higher frequencies approaching 1 MHz, a noticeable reduction in gain was observed.

This reduction occurs due to the finite bandwidth and gain-bandwidth product of the operational amplifier, as well as internal capacitances that limit high-frequency performance.

Conclusion.

The results demonstrate that a non-inverting amplifier maintains a stable gain at low and mid frequencies but experiences gain roll-off at higher frequencies. This behaviour is consistent with op-amp theory and confirms that frequency response is a key limitation in practical amplifier design.

Simulated Non-Inverting Amplifier - Effects of Changing the Frequency on Gain Results Table			
Frequency (Hz)	V _{in} (mV)	V _{out} (mV)	V _{Gain} (A _v) = V _{out} /V _{in}
1K	200mV	639mV	3.195
10K	200mV	639mV	3.195
100K	200mV	607mV	3
500K	200mV	293mV	1.5
1M	200mV	154mV	0.77

Before carrying out the next task, read through & complete the electrical safety tick sheet below. Ensure that you work safely at all times.

IF IN DOUBT ASK.

Safety checklist	Check acknowledged
Plan the electronic assembly, wiring and testing activities before you start them.	✓
Use appropriate sources to obtain the required specifications, circuit diagrams, component assembly and test information	✓
Obtain the correct PPE, tools & equipment for the assembly and test operations. Check that they are in a safe and usable condition.	✓
Use the appropriate methods and techniques to assemble the components in their correct positions	✓

Unit 19: Electronic Devices and Circuits

Assignment A

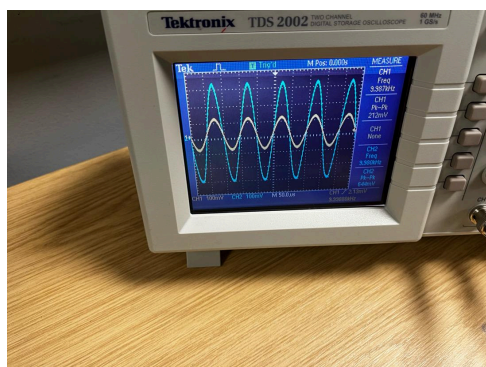
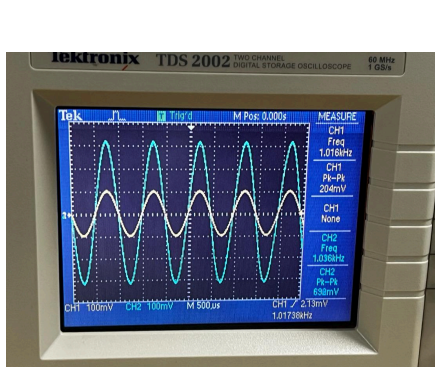
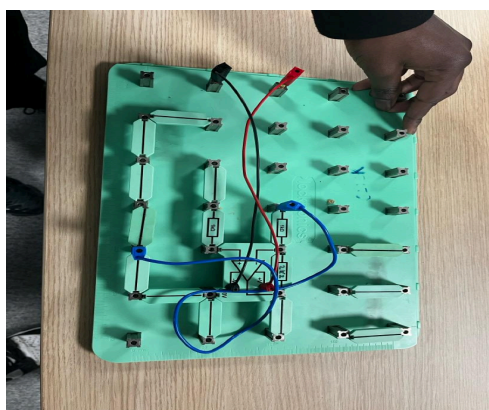
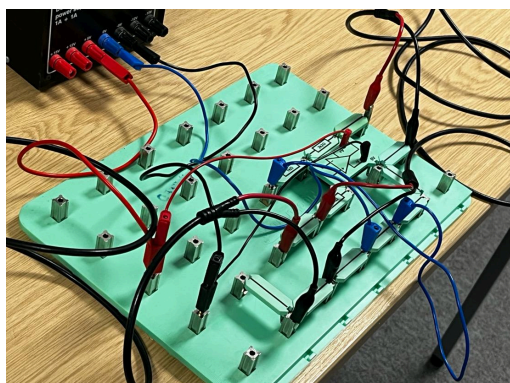
Ensure that power is isolated when making adjustments to the circuits	✓
Assemble and secure the components, using the specified connectors, securing devices & appropriate techniques.	✓
Ensure wire and cables are connected to the appropriate connections on the circuit boards.	✓
Ensure that you use appropriate test methods and equipment to check that the completed assembly is safe and meets all aspects of the specification.	✓
Ensure the work area is in a safe and tidy condition during & on completion of the electronic assembly and testing activities	✓
Ensure that all polarity sensitive components are correctly installed.	✓

Unit 19: Electronic Devices and Circuits

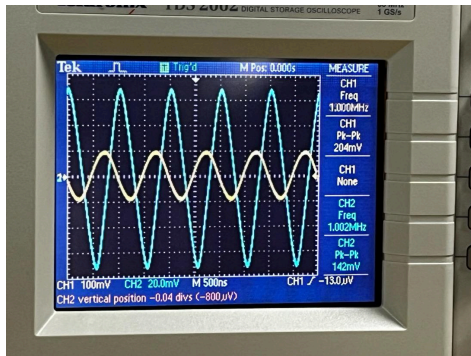
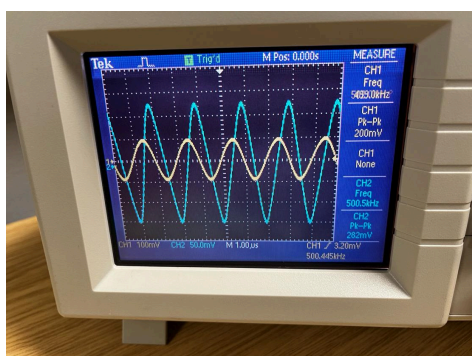
Assignment A

(v) build a **real-life prototype** circuit and measure the **voltage gain** over the same range of frequencies.

Provide image evidence of your practical set-up & provide details of your findings. Record your results in the table provided using the given formula.

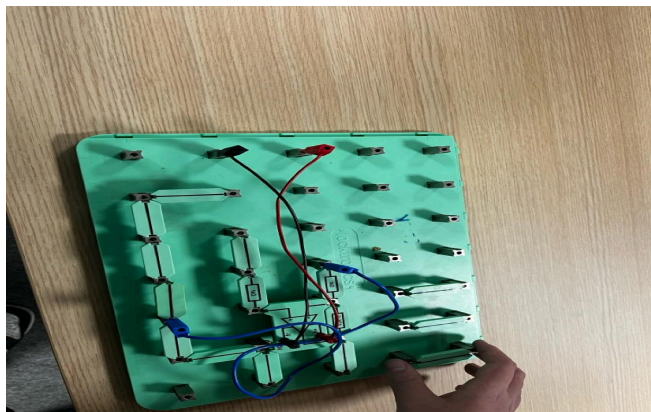


*pictures of the practical simulations done by me.



values obtained from the practicals.

*the various



* image evidence of practicals.

Practical investigation on frequency effects on voltage gain.

A real-life prototype of the non-inverting amplifier circuit was constructed using the same component values as the simulated design. The circuit was assembled on a breadboard and powered using a dual DC power supply. A function generator was used to apply a sine wave input signal of constant amplitude, while the frequency was varied from 1 kHz to 1 MHz.

An oscilloscope was used to measure both the input voltage (V_{in}) and output voltage (V_{out}) at each frequency. The voltage gain was calculated using the given formula: $A_v = V_{out} / V_{in}$

Findings.

At low frequencies, the measured voltage gain closely matched both the theoretical and simulated values, confirming correct circuit construction and biasing. As the frequency increased, the gain remained stable through the mid-frequency range but gradually decreased at higher frequencies. The reduction in gain became more noticeable near the upper end of the frequency range due to op-amp bandwidth limitations, parasitic capacitance, and component tolerances present in the physical circuit.

Conclusion.

The practical results follow the same trend observed in simulation, demonstrating

that non-inverting amplifier gain is frequency-dependent in real-world applications. Minor differences between simulated and measured values are expected and confirm the importance of practical testing alongside theoretical and simulation-based analysis.

Practical Non-Inverting Amplifier -Effects of Changing the Frequency on Gain Results Table			
Frequency (Hz)	Vin (mV)	Vout (mV)	VGain (Av) = Vout/Vin
1K	200mV	640mV	3.2
10K	200mV	600mV	3
100K	200mV	572mV	2.9
500K	200mV	282mV	1.41
1M	200mV	142mV	0.71

(vi) Compare the results from **theory, simulation & measurement** and provide an overall conclusion of your findings through research.

Comparison and overall conclusion.

This investigation combined theoretical analysis, circuit simulation, and practical measurement to evaluate the performance of analogue amplifier circuits over a range of operating conditions. Using all three approaches provided a comprehensive understanding of circuit behaviour and highlighted the strengths and limitations of each method.

Theoretical results.

Theoretical calculations were carried out using standard amplifier gain equations for non-inverting and inverting operational amplifier circuits. These calculations assume ideal components, meaning no internal resistance, no bandwidth limitations, and perfectly linear behaviour.

From theory; the calculated voltage gain remained constant across all frequencies, the predicted gain solely depended on resistor ratios and phase behavior matched expected outcomes(inverting and non-inverting).

While theory provides an essential foundation for circuit design, it does not account for real-world effects such as parasitic capacitance, internal op-amp limitations, or

component tolerances. As a result, theoretical values consistently produced the highest gain values.

Simulation results.

Simulation using Multisim introduced non-ideal component models, including realistic op-amp parameters such as finite gain-bandwidth product and internal capacitances. From simulation; gain values at low and mid frequencies matched theoretical predictions and phase behavior and waveform integrity were accurately represented.

Simulation provided a more realistic representation of circuit behaviour compared to theory, while still offering a controlled and repeatable testing environment. Minor deviations from theory confirmed the impact of real device limitations.

Practical results.

The practical prototype allowed direct measurement of circuit behaviour using real components and test equipment. These results showed the most variation due to unavoidable real-world influences. From practical testing; measured value gains were slightly lower than both theoretical and simulated, gain reduction at high frequencies occurred quite early and was noticeable and the overall results are affected by component tolerances, breadboard parasitic capacitance, wiring resistance, and instrument limitations.

Despite these factors, the overall trend closely matched both theoretical predictions and simulation results, confirming the validity of the design.

Comparative evaluation.

Aspect	Theory	Simulation	Practical
Accuracy	Ideal	High	Moderate
Realism	Low	High	Very High
Frequency effects	Not shown	Clearly shown	Strongly evident
Usefulness	Design stage	Validation	Final confirmation

Theoretical calculations were essential for initial design, simulation was critical for predicting realistic behaviour, and practical testing was necessary for verifying real-world performance.

OVERALL CONCLUSION.

In conclusion, the investigation demonstrated that analogue amplifier performance is best understood by combining theory, simulation, and practical testing. While theoretical calculations provide accurate baseline expectations, simulation reveals

realistic operating limits, and practical measurements confirm real-world behaviour. The close agreement across all three methods validates the circuit designs and confirms that operational amplifier gain is controlled by resistor values but limited by bandwidth and physical construction. This multi-method approach is essential in professional electronic engineering practice and ensures safe, reliable, and predictable circuit operation.

REFERENCE LISTS.

I sourced info from the following sites:

- **electronics-tutorials .ws** (for info about zener diodes and its behaviours)
- [facebook.com](https://www.facebook.com) (for info about general electronics and its functions)
- askfilo.com (for info about theoretical and practical findings)
- anypcba.com (for info about some electronic terms)
- analog.com (for info about diagrams and basic electronics)